

深度學習
Deep Learning

⭐ DL Training Tips

Instructor: 林英嘉 (Ying-Jia Lin)
2026/10/21

Course GitHub

Slido # DL1021

Outline

- Model training tips [70 min]
  - EarlyStopping
  - Regularization and AdamW
  - Batch Normalization
  - Dropout
- Code demo [10 min]
- Term Project [50 min]
- Quiz [20 min]



[Recap] 如何訓練模型：資料集

Training Set
(訓練資料集)



Test Set
(測試資料集)



Training Set
(訓練資料集)

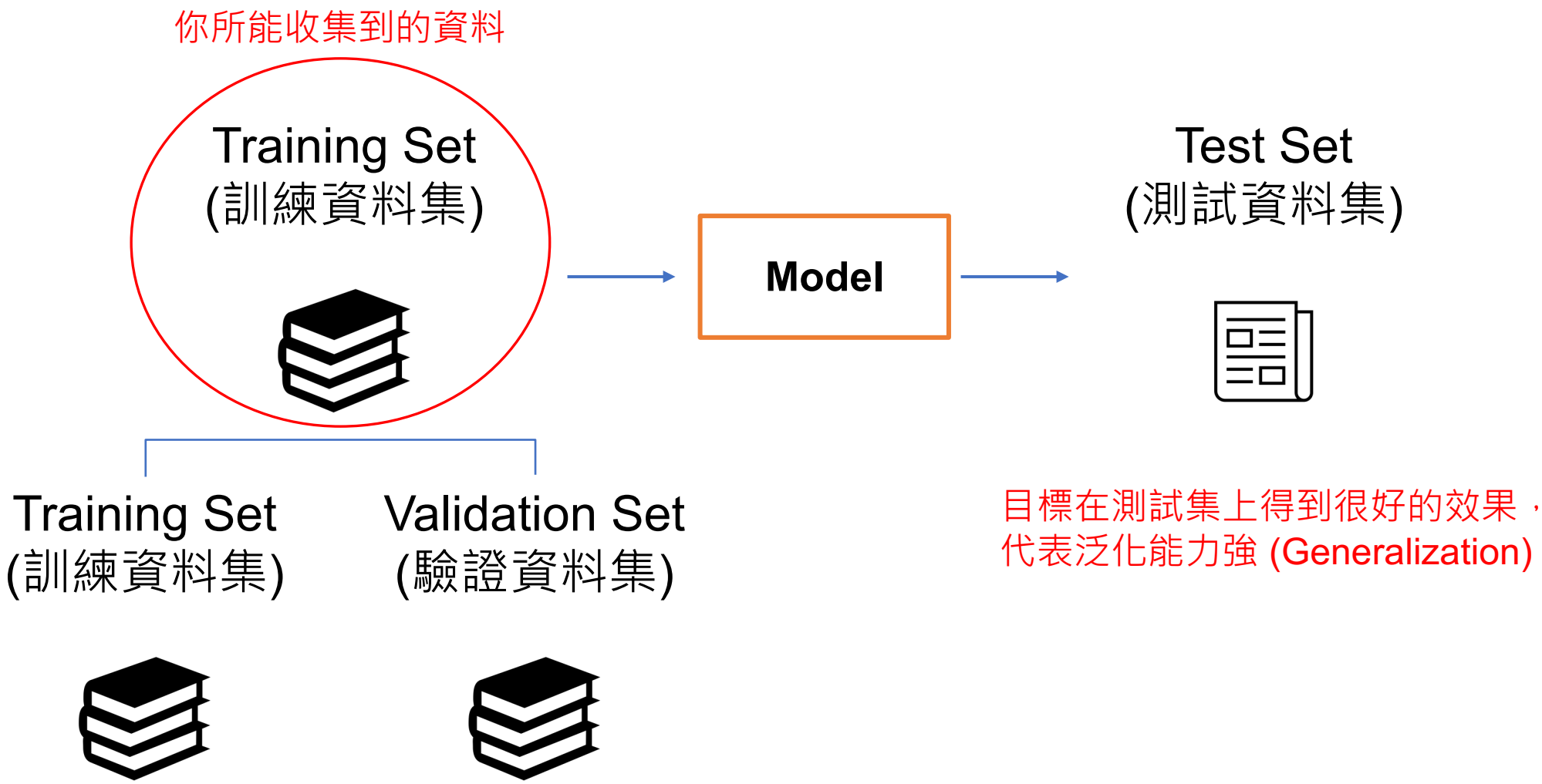


Validation Set
(驗證資料集)



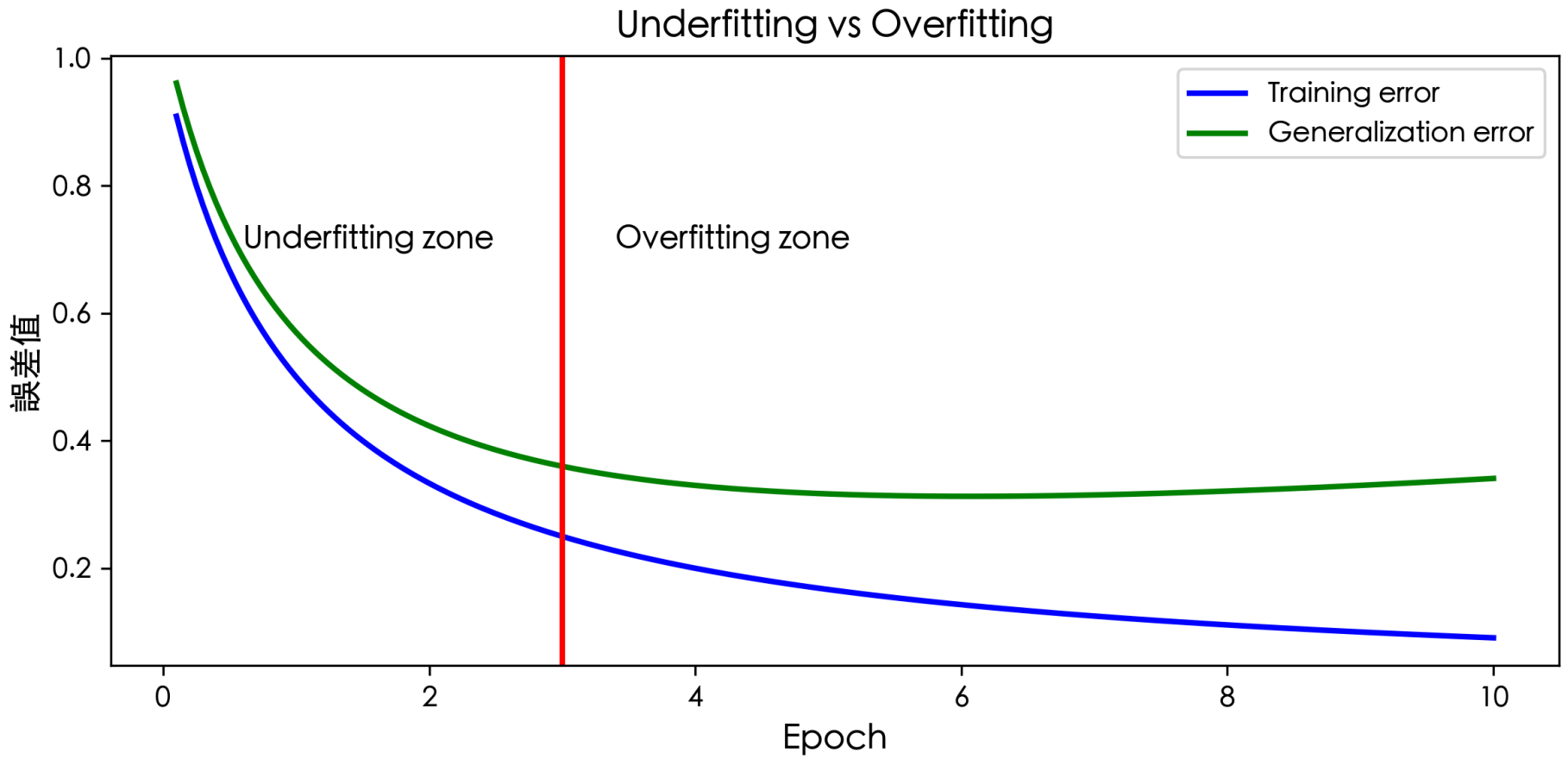


訓練模型的目標


你所能收集到的資料
Training Set
(訓練資料集)
Model
Test Set
(測試資料集)
Training Set
(訓練資料集)
Validation Set
(驗證資料集)
目標在測試集上得到很好的效果，
代表泛化能力強 (Generalization)

訓練機器學習模型時可能發生的情況

code/w7_plot_overfitting.py





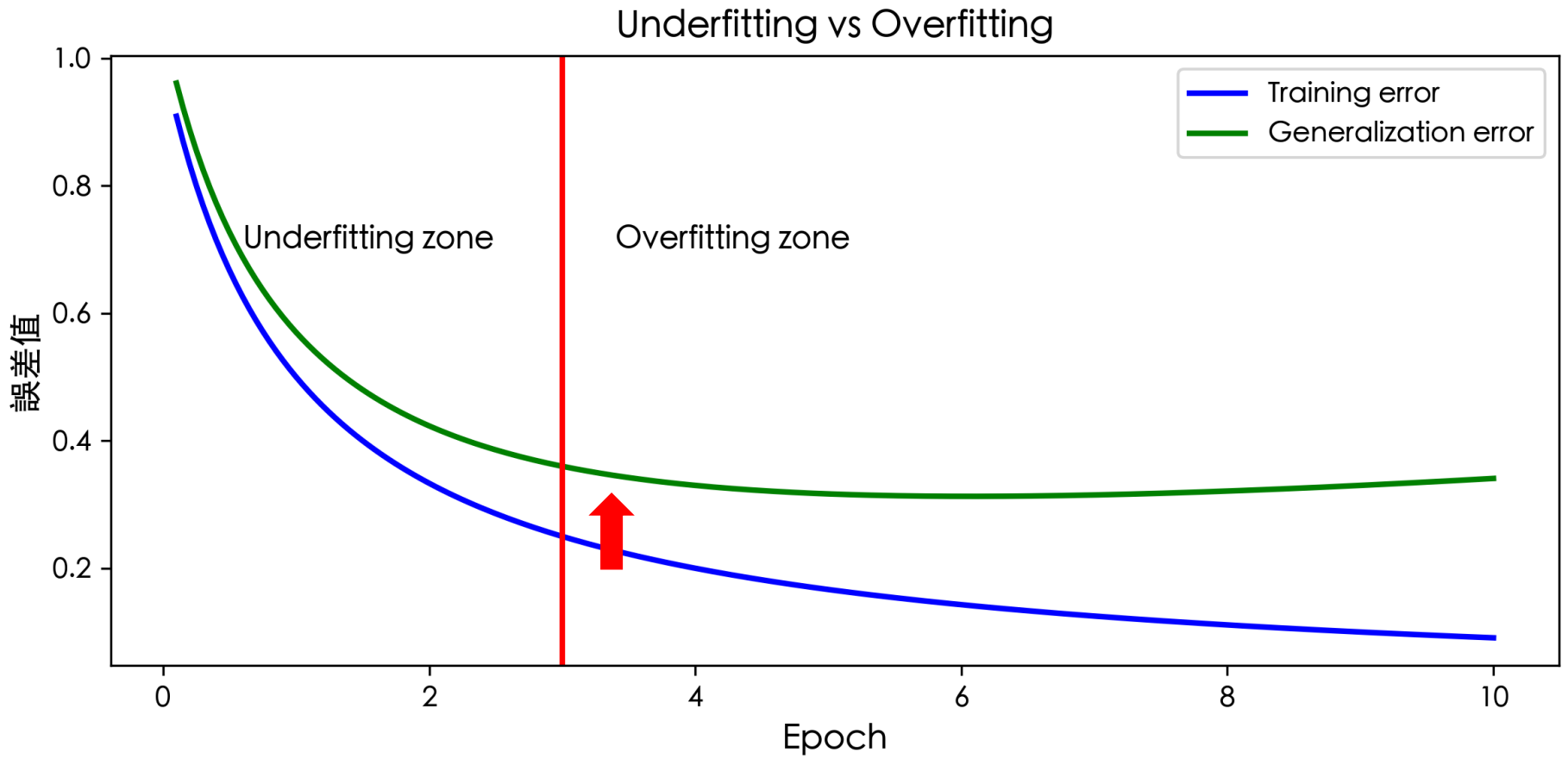
[定義] Overfitting 與 Underfitting

- Underfitting (欠擬合): 模型無法在 training set 取得很低的誤差 (error; loss)
- Overfitting (過擬合): 模型的 training error 與 test error 差異很大
  - test error 通常代表 validation loss
  - 常見模型可以呈現很小的 training error，但 test error 仍大



Underfitting vs. Overfitting

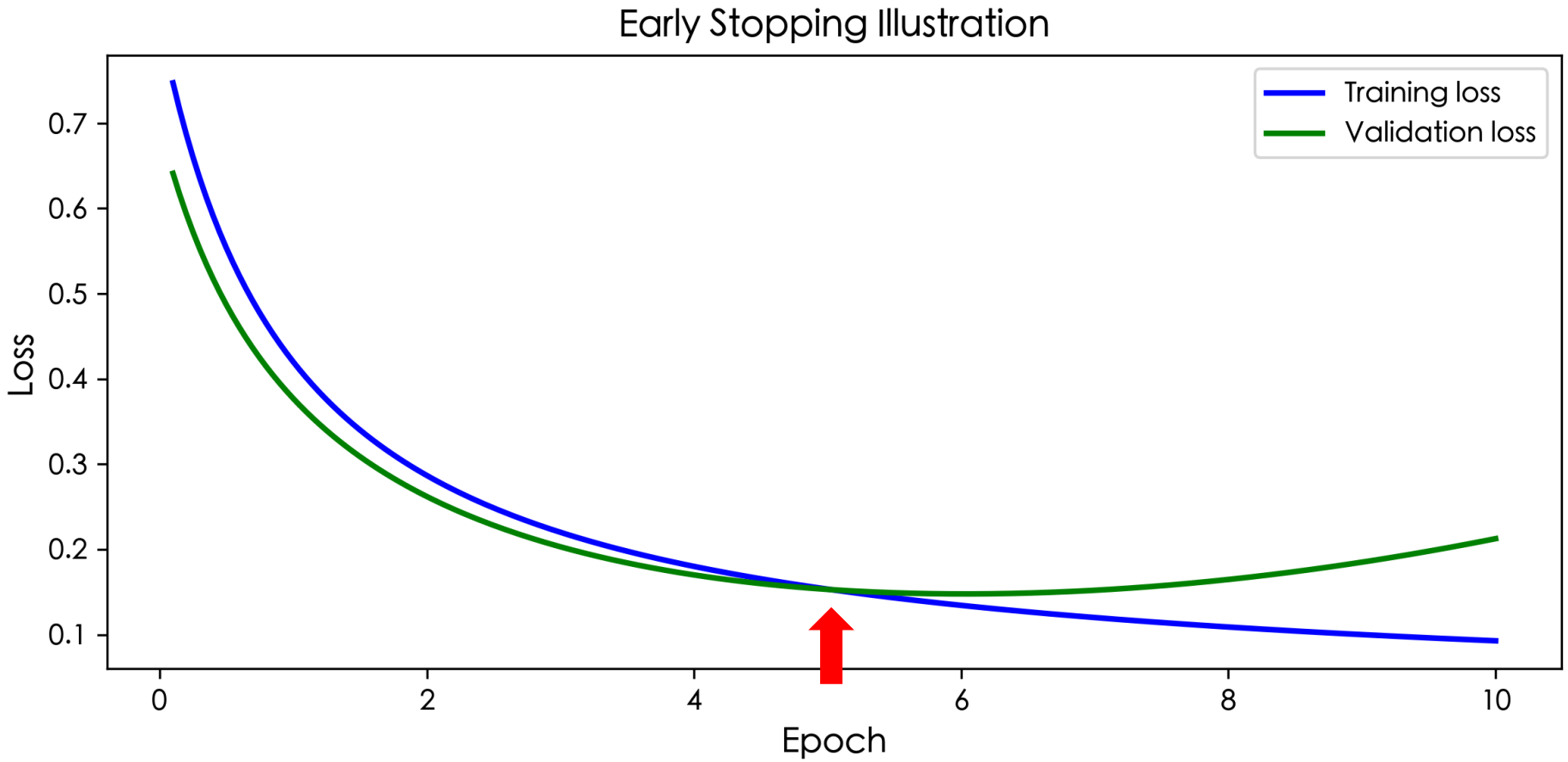
code/w7_plot_overfitting.py





EarlyStopping 的良好時間點

code/w7_plot_es.py





Regularization

- 定義：regularization (正則化) 是一種技巧來幫助模型減少 training error 與 test error 的差距 (並非降低 training error)

$$\mathcal{L}_{\text{new}} = \mathcal{L} + \boxed{\text{regularization_term}}$$



Norm (範數)

是具有「長度」概念的函數，其為向量空間內的所有向量的正長度或大小。

假設有一向量 $\mathrm{w} = [w_1, w_2, \ldots, w_n]$

$$\text{L}^1\text{-Norm} = \|\mathrm{w}\|_1 = \sum_{i=1}^{n} |w_i| = |w_1| + |w_2| + \cdots + |w_n|$$

$$\text{L}^2\text{-Norm} = \|\mathrm{w}\|_2 = \sqrt{w_1^2 + w_2^2 + \cdots + w_n^2}$$



L^2 Regularization

$\mathrm{w} = \{w_1, w_2, \ldots, w_n\}$
λ: 正則項的強度 (0-1之間)
* $\frac{1}{2}$ 是慣例，可不加

$$\mathcal{L}_{\text{new}} = \mathcal{L} + \lambda \frac{1}{2} \|\mathrm{w}\|_2^2$$

$$w_1^2 + w_2^2 + \cdots + w_n^2$$

代表模型的目標除了要讓 $\mathcal{L}$ 越小越好之外，
還要讓所有的 w 的平方和越小越好



L^2-Norm 說明：https://mathworld.wolfram.com/L2-Norm.html

為什麼要增加正則項？

假設我們的模型：$\hat{y} = x_1 w_1 + x_2 w_2^2 + \cdots + x_9 w_9^9$

如果沒有正則項，目標函數相當於
是為 training set 量身打造

如果 w_9很大怎麼辦？
正則項 (如 L^2) 能限制
weight 數值大小

假設我們想要學出的模型：$y = x_1 w_1 + x_2 w_2^2$

非常通用的模型，
可以在 unseen 資料上表現好



Gradient Descent with L^2 Regularization

$$\mathcal{L}_{\text{new}} = \mathcal{L} + \lambda \frac{1}{2} \|\mathrm{w}\|_2^2$$

對任意 w_i 進行偏微分：

$$\frac{\partial \mathcal{L}_{\text{new}}}{\partial w_i} = \boxed{\frac{\partial \mathcal{L}}{\partial w_i} + \lambda w_i}$$

梯度下降：

$$w_i' = w_i - \eta \frac{\partial \mathcal{L}_{\text{new}}}{\partial w_i}$$

(代入) $$= w_i - \eta\left(\frac{\partial \mathcal{L}}{\partial w_i} + \lambda w_i\right)$$

(移項) $$= \underline{(1 - \eta\lambda)} w_i - \eta\left(\frac{\partial \mathcal{L}}{\partial w_i}\right)$$

每次更新都會倍數遞減
(weight decay)



L^1 Regularization

$\mathrm{w} = \{w_1, w_2, \ldots, w_n\}$
λ: 正則項的強度 (0-1之間)

$$\mathcal{L}_{\text{new}} = \mathcal{L} + \lambda \|\mathrm{w}\|_1$$

$$\text{L}^1\text{-Norm} = \sum_{i=1}^{n} |w_i| = |w_1| + |w_2| + \cdots + |w_n|$$

代表模型的目標除了要讓 $\mathcal{L}$ 越小越好之外，
還要讓所有的 w 的絕對值和越小越好

L¹-Norm 說明：https://mathworld.wolfram.com/L1-Norm.html

Gradient Descent with L1 Regularization

$$\mathcal{L}_{\text{new}} = \mathcal{L} + \lambda \|\mathrm{w}\|_1$$

對任意 w_i 進行偏微分：

$$\frac{\partial \mathcal{L}_{\text{new}}}{\partial w_i} = \frac{\partial \mathcal{L}}{\partial w_i} + \lambda \cdot \mathrm{sign}(w_i)$$

梯度下降：

$$w_i' = w_i - \eta \frac{\partial \mathcal{L}_{\text{new}}}{\partial w_i}$$

(代入) $$= w_i - \eta\left(\frac{\partial \mathcal{L}}{\partial w_i} + \lambda \cdot \mathrm{sign}(w_i)\right)$$

每次更新都會扣減
(weight decay)



AdamW will be in Week 7

The paper of AdamW:

Published as a conference paper at ICLR 2019

DECOUPLED WEIGHT DECAY REGULARIZATION

Ilya Loshchilov & Frank Hutter
University of Freiburg
Freiburg, Germany,
{ilya,fh}@cs.uni-freiburg.de

Week7: 過擬合、正規化、模型訓練技巧、期末專案介紹



[Recap] Adam

- Adam: adaptive moment estimation
- 結合了 Momentum、Adagrad 和 RMSProp 的概念



Kingma, Diederik P., and Jimmy Ba. "Adam: A method for stochastic optimization." ICLR 2015. https://arxiv.org/abs/1412.6980

[Recap] 如何有效訓練深度學習模型？

Optimizer:

- 對梯度動手腳
  - Momentum
- 自動調整學習率
  - Adagrad
  - RMSprop
  - Adam



[Recap] Adam 流程

- Adam: Momentum、Adagrad 和 RMSProp 的概念

1. 計算梯度 $g_t \leftarrow \nabla_w f(w_{t-1})$
2. 以移動平均紀錄梯度 g_t (為了對梯度動手腳 -- Momentum)
3. 以移動平均紀錄梯度平方 g_t^2 (為了調整學習率 -- Adagrad)
4. 更新參數 $w_t \leftarrow w_{t-1} - \eta \hat{m}_t \frac{1}{\sqrt{\hat{v}_t + \epsilon}}$



Kingma, Diederik P., and Jimmy Ba. "Adam: A method for stochastic optimization." ICLR 2015. https://arxiv.org/abs/1412.6980

[實作細節] 把 L^2 Regularization 加到 SGD

總目標： $\mathcal{L}_{\text{new}} = \mathcal{L} + \lambda \frac{1}{2} \|\mathrm{w}\|_2^2$

Loshchilov, Ilya, and Frank Hutter. “Decoupled Weight Decay Regularization.” International Conference on Learning Representations (2019).

1. 計算梯度 $g_t \leftarrow \nabla_{\mathrm{w}} f(\mathrm{w}_{t-1}) + \boxed{\lambda \mathrm{w}_{t-1}}$

請見 AdamW 論文 Proposition 1 proof
或本簡報 p13

2. 更新參數 $w_t \leftarrow w_{t-1} - \eta g_t$

等價於

GD： $w_t = w_{t-1} - \eta(\frac{\partial \mathcal{L}}{\partial w_{t-1}} + \lambda w_{t-1})$



[實作細節] 把 L2 Regularization 加到 Adam

在SGD正確，但在Adam是錯的，因為改變1的梯度會影響到梯度紀錄

1. 計算梯度 $g_t \leftarrow \nabla_w f(w_{t-1}) + \lambda w_{t-1}$
2. 以移動平均紀錄梯度 g_t (為了對梯度動手腳 -- Momentum)
3. 以移動平均紀錄梯度平方 g_t^2 (為了調整學習率 -- Adagrad)
4. 更新參數 $w_t \leftarrow w_{t-1} - \eta \hat{m}_t \frac{1}{\sqrt{\hat{v}_t} + \epsilon}$



Loshchilov, Ilya, and Frank Hutter. “Decoupled Weight Decay Regularization.” International Conference on Learning Representations (2019).

AdamW

分離的

AdamW: Adam with decoupled weight decay

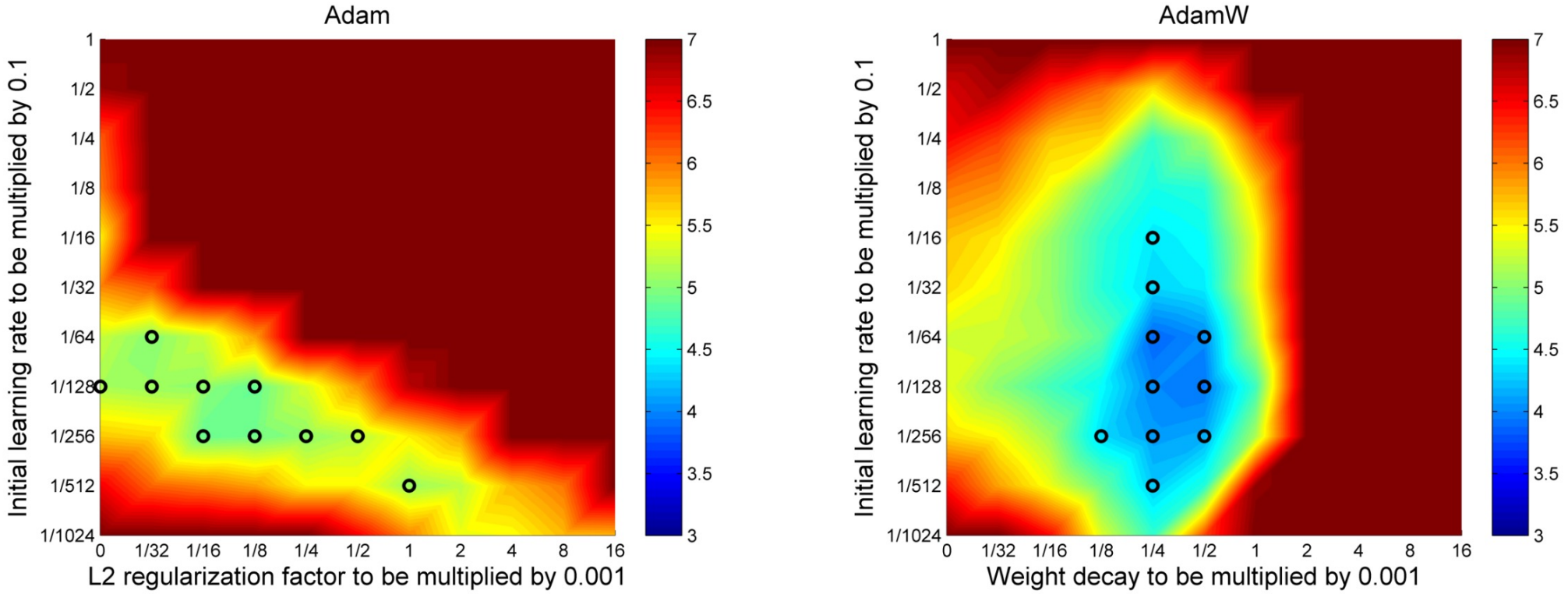
1. 計算梯度 $g_t \leftarrow \nabla_w f(w_{t-1}) + \lambda w_{t-1}$
2. 以移動平均紀錄梯度 g_t (為了對梯度動手腳 -- Momentum)
3. 以移動平均紀錄梯度平方 g_t^2 (為了調整學習率 -- Adagrad)
4. 更新參數 $w_t \leftarrow w_{t-1} - \eta(\hat{m}_t \frac{1}{\sqrt{\hat{v}_t}+\epsilon} + \lambda w_{t-1})$

$$w_t = w_{t-1} - \eta(\frac{\partial \mathcal{L}}{\partial w_{t-1}} + \lambda w_{t-1})$$



Loshchilov, Ilya, and Frank Hutter. “Decoupled Weight Decay Regularization.” International Conference on Learning Representations (2019).

Performance: Adam vs. AdamW



Loshchilov, Ilya, and Frank Hutter. “Decoupled Weight Decay Regularization.” International Conference on Learning Representations (2019).

延伸學習

李宏毅老師影片

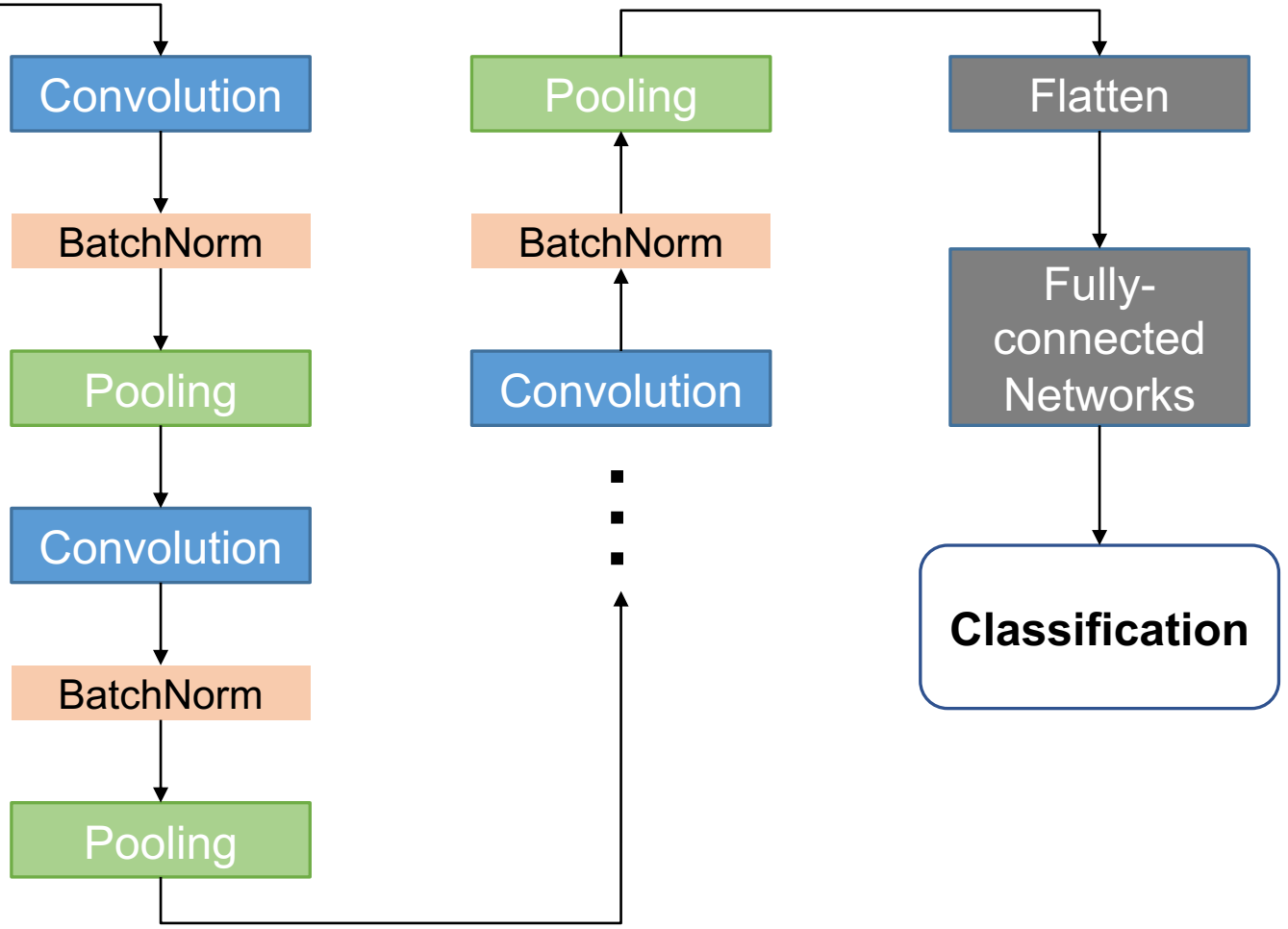
https://youtu.be/xki61j7z-30?si=ofWigNuHKkTdPC4f&t=3463



Batch Normalization

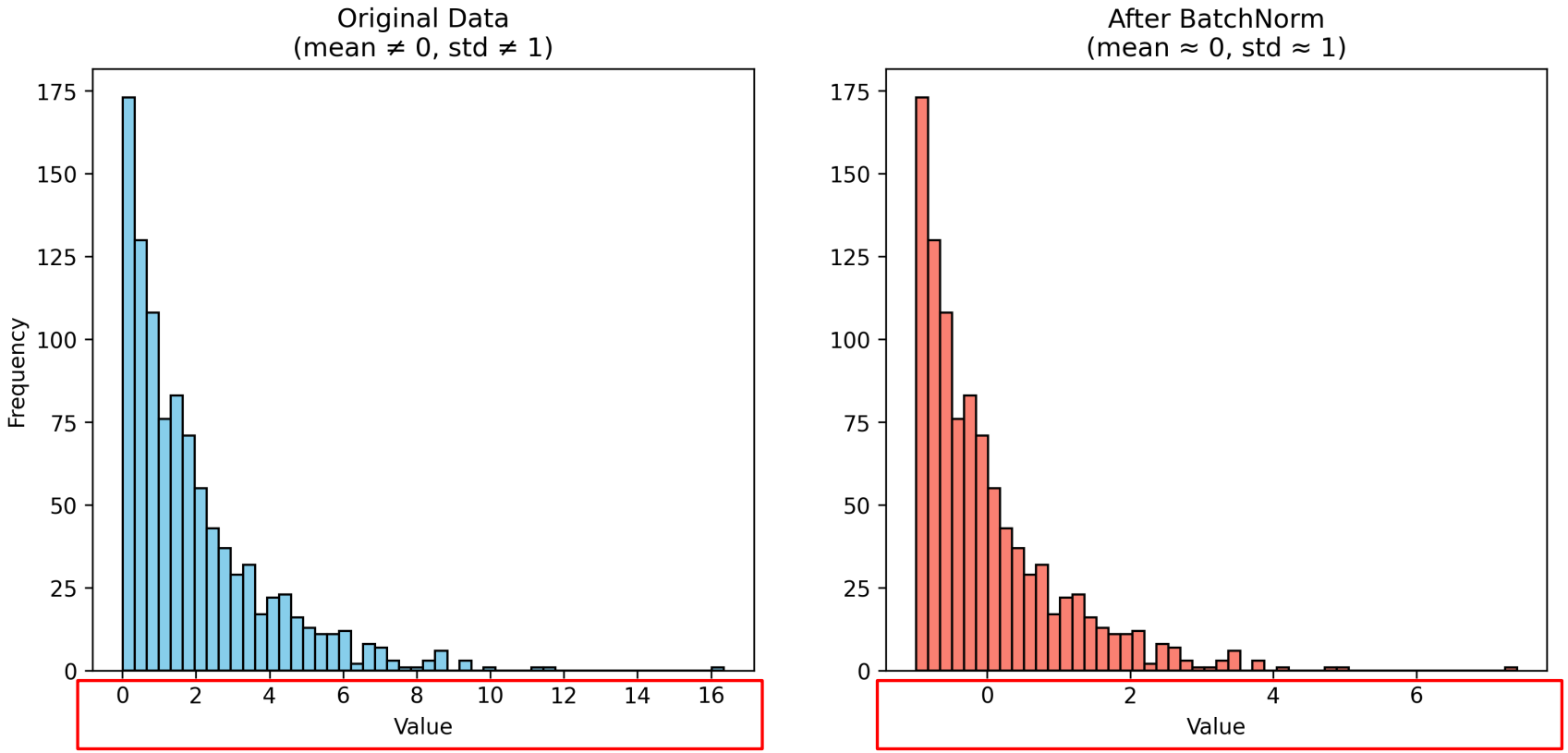
The whole CNN



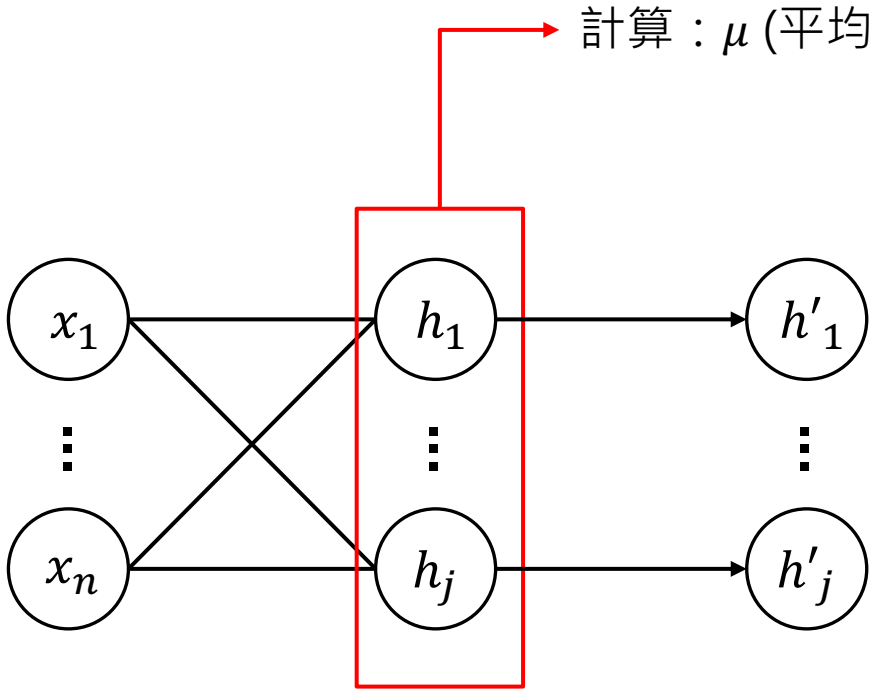


Batch Normalization (BatchNorm)

- BatchNorm 對數值進行轉換，使得轉換後的數值的平均值為零、標準差為1
- 類似於統計上的 Z-score standardization



MLP 中的 BatchNorm

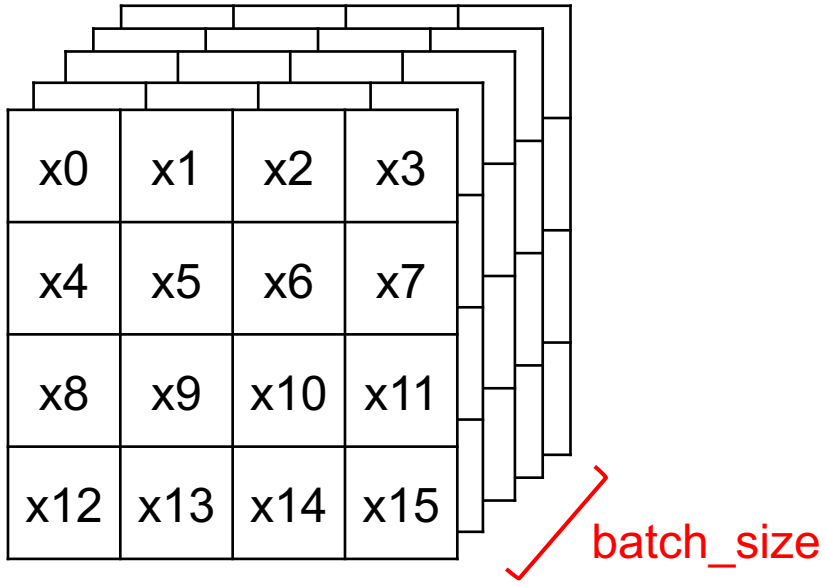


$$h'_1 = \frac{h_1 - \mu}{\sigma}$$

$$h'_j = \frac{h_j - \mu}{\sigma}$$

Ioffe, Sergey, and Christian Szegedy. "Batch normalization: Accelerating deep network training by reducing internal covariate shift." International conference on machine learning. 2015.

CNN 中的 BatchNorm



(5, 4, 4)

(batch_size, height, width)

μ 計算自這個 batch 的每一個值

σ 計算自這個 batch 的每一個值

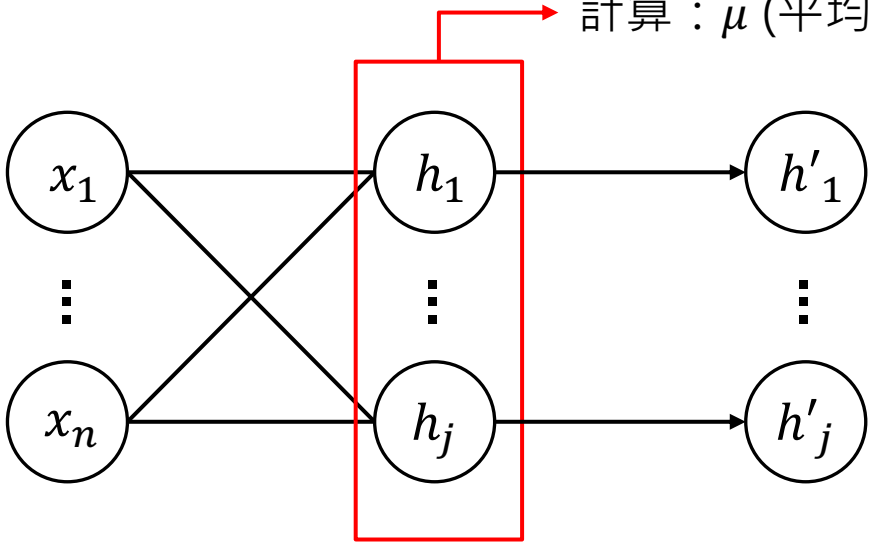


參考：https://stackoverow.com/questions/65613694/calculation-of-mean-and-variance-in-flbatch-normalization-in-convolutional-neural

BatchNorm During Test (Inference)

Training

計算：μ (平均數) & σ (標準差)



https://pytorch.org/docs/stable/generated/torch.nn.BatchNorm2d.html

Moving average

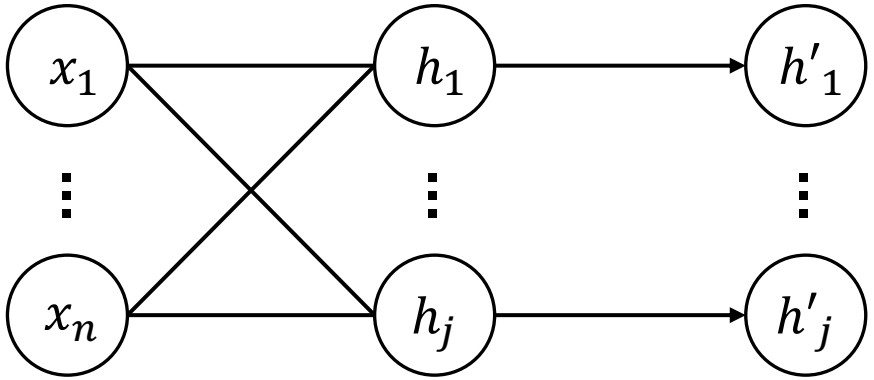
$$\mu_t = \alpha\mu_t + (1-\alpha)\mu_{t-1}$$

$$\sigma_t = \alpha\sigma_t + (1-\alpha)\sigma_{t-1}$$

momentum (α) 常見設定為0.1

Inference

假設只有1張影像



$$h'_1 = \frac{h_1 - \mu_t}{\sigma_t}$$

$$h'_j = \frac{h_j - \mu_t}{\sigma_t}$$

[觀察] BatchNorm

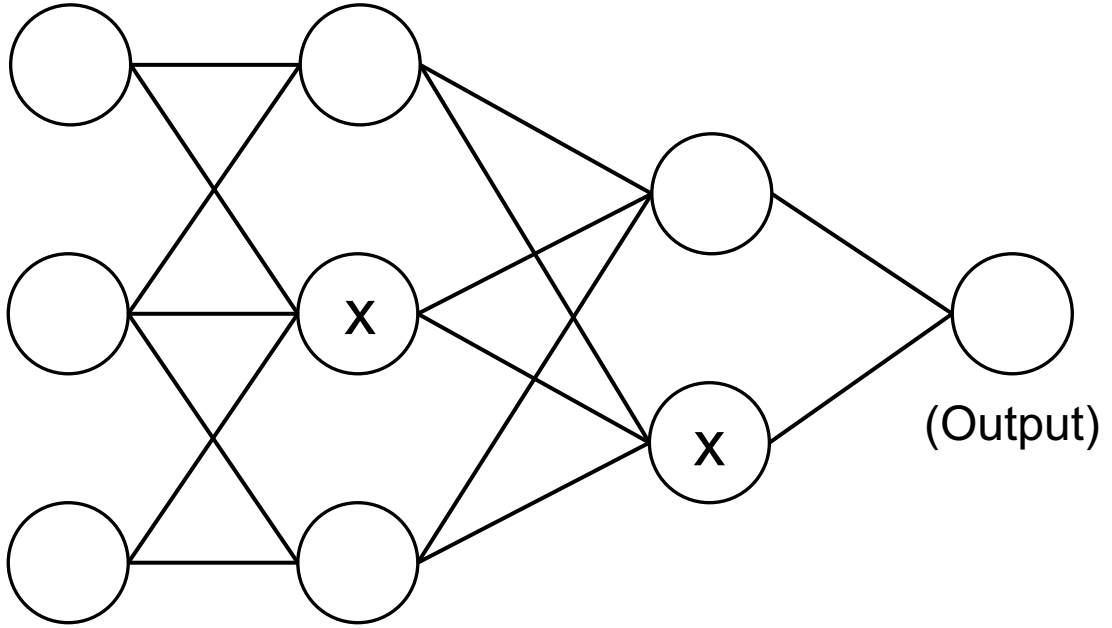
- BatchNorm 可以：
  - 減少不同特徵尺度帶來的偏差，避免梯度爆炸或消失
  - 提高模型訓練時的穩定性，加速收斂



Dropout

Dropout

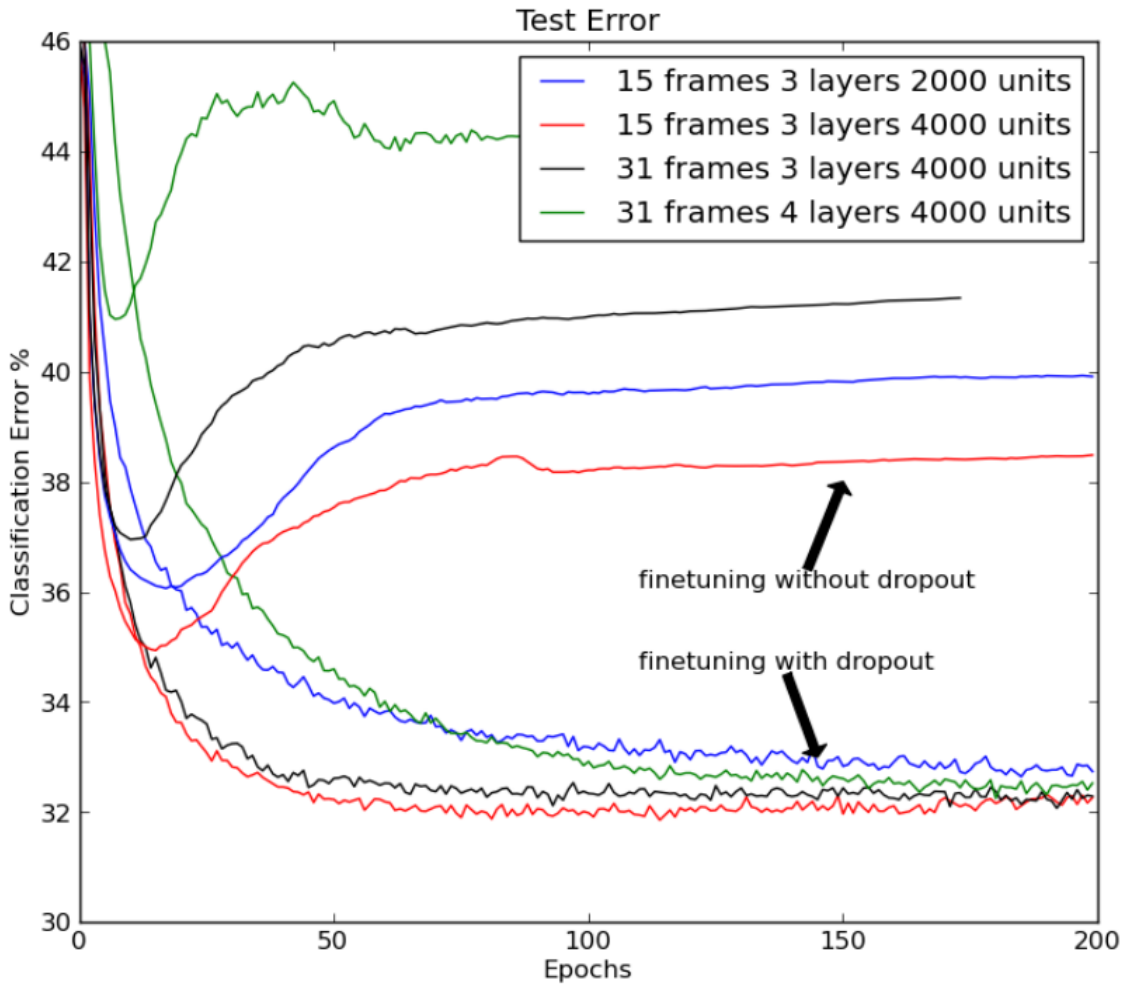
- 每輪訓練時，神經網路每一層的輸入值部份以 p 的機率隨機替換成 0
  - p 是超參數
- 可防止模型過擬合 (over-fitting)，增強模型的泛化能力 (generalization)





Srivastava, Nitish, et al. "Dropout: a simple way to prevent neural networks from overfitting." The journal of machine learning research 15.1 (2014): 1929-1958.

Dropout may improve generalization



Srivastava, Nitish, et al. "Dropout: a simple way to prevent neural networks from overfitting." The journal of machine learning research 15.1 (2014): 1929-1958.



model.eval()

- 使用 training 時取得的平均值跟標準差來算 batch normalization
- 關閉 dropout (保留所有的神經元)
  - 測試時已經不需要再更新模型
  - 去除隨機性



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Project Introduction

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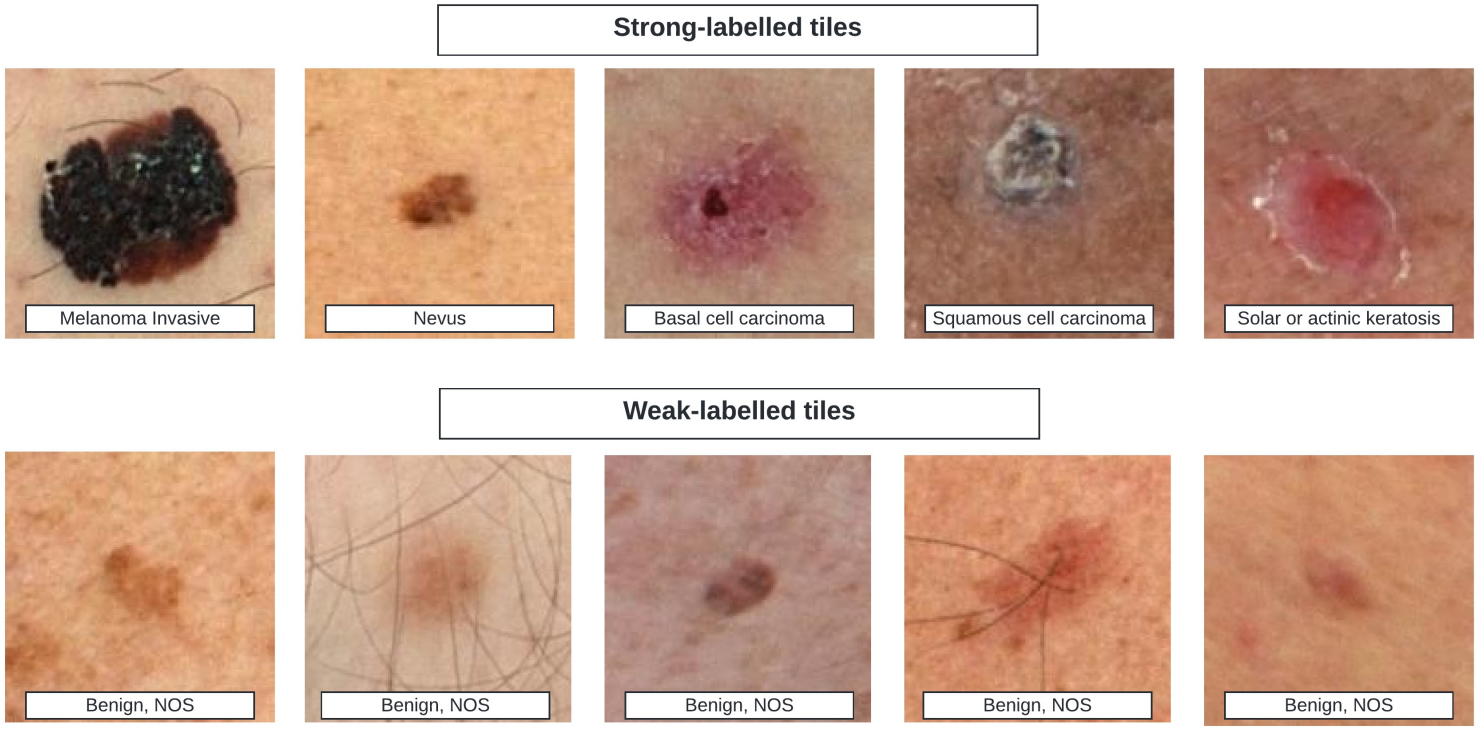
Outline of tasks

Platform	Competition Name	資料類型	難度	任務類型	特別之處
Kaggle	ISIC 2024 - Skin Cancer Detection with 3D-TBP	Image	⭐	Binary Classification*	Class imbalance
	RSNA Pneumonia	Image	⭐⭐	Object Detection	非單純分類
	Name That LLM	Text	⭐	Text Classification*	資料分佈
Codabench	SemEval 2027 Task 2	Text	⭐⭐	Text Classification	範例較少



* Code Competition

ISIC 2024 - Skin Cancer Detection with 3D-TBP



test-metadata & submission.csv

test-metadata.csv (2.21 kB)

Detail Compact Column 10 of 44 columns

isic_id	patient_id	age_approx
3 unique values	3 unique values	3 total v
ISIC_0015657	IP_6074337	45.0
ISIC_0015729	IP_1664139	35.0
ISIC_0015740	IP_7142616	65.0

No more data to show

Data Explorer

2.69 GB

- train-image
- sample_submission.csv
- test-image.hdf5
- test-metadata.csv
- train-image.hdf5
- train-metadata.csv

Summary

- 401k files
- 101 columns

sample_submission.csv (66 ...

Detail Compact Column 2 of 2 columns

isic_id	target
3 unique values	3 total values
ISIC_0015657	0.3
ISIC_0015729	0.3
ISIC_0015740	0.3

No more data to show



ROC Curve

(Useful!!) https://developers.google.com/machine-learning/crash-course/classification/roc-and-auc?hl=zh-tw

ROC curve: Receiver operating characteristic curve (接收者操作特徵曲線)

ROC curve 定義: 針對不同的模型分類門檻下的模型表現圖

- X軸：False positive rate (FPR); Y軸：True positive rate (TPR)
- AUC: ROC 的底面下面積 (越大越好)

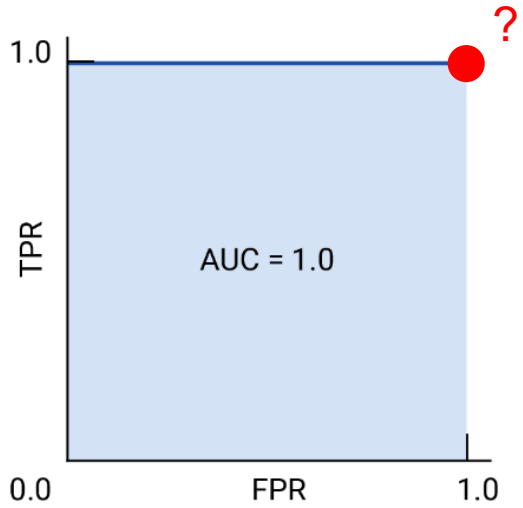


圖 1：ROC 和 AUC，這是一個完美假設的模型。

模型分類門檻 (Threshold)



Model

Rust

模型輸出	答案
機率 = 0.55	1

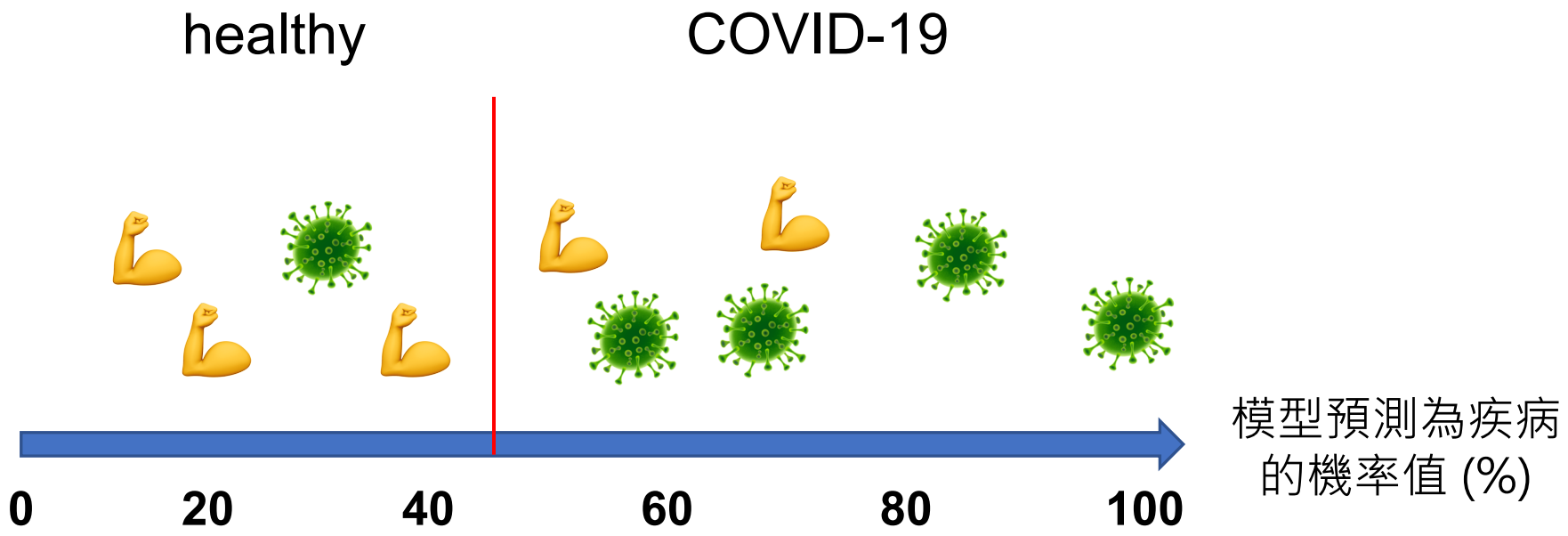
我們應該設多少門檻？

threshold = 0.5 -> 模型預測為1，正確數量+1

threshold = 0.6 -> 模型預測為0，正確數量-1

TPR and FPR 範例

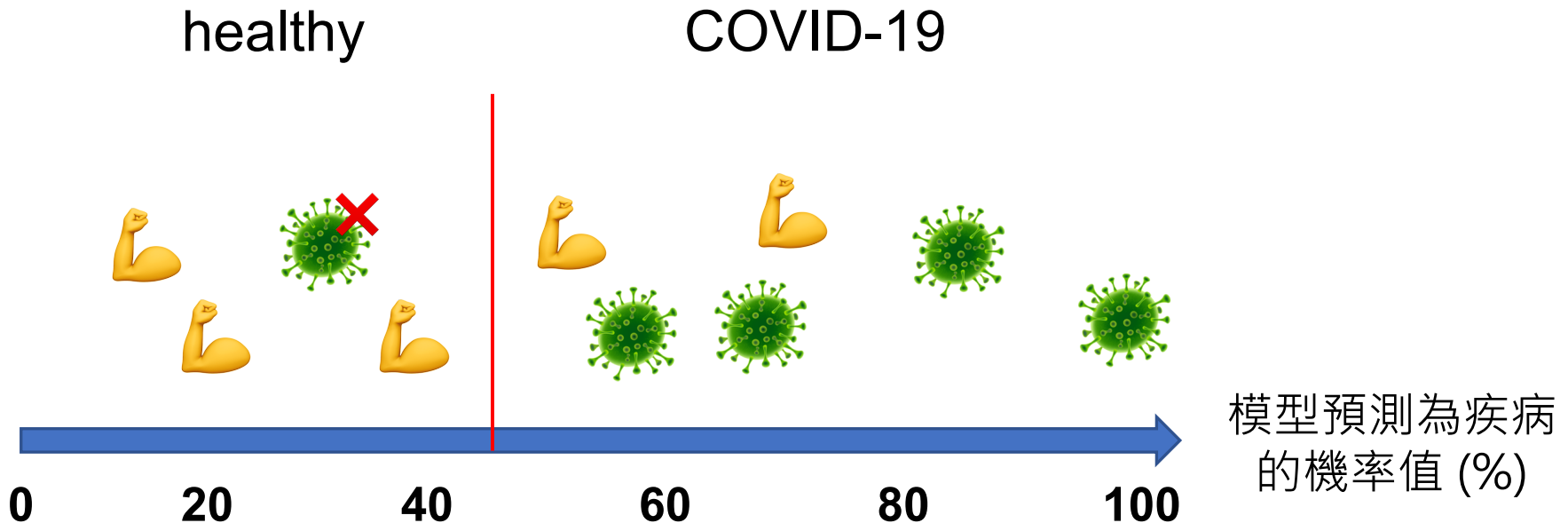
假設我們用模型對10個人做COVID-19檢測：
(COVID-19 代表 positive)



TPR and FPR 範例 (TPR)

假設我們用模型對10個人做COVID-19檢測：
(healthy 代表 negative; COVID-19 代表 positive)

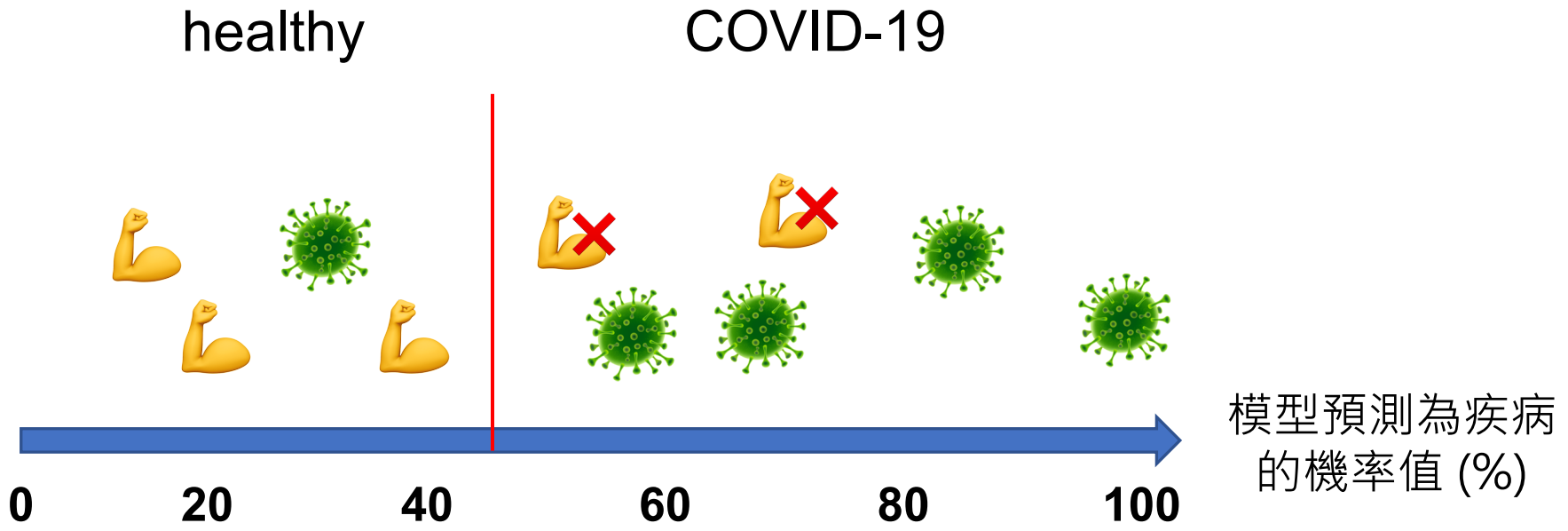
True positive rate
(TPR) = 4 / 5 = 0.8



TPR and FPR 範例 (FPR)

假設我們用模型對10個人做COVID-19檢測：
(healthy 代表 negative; COVID-19 代表 positive)

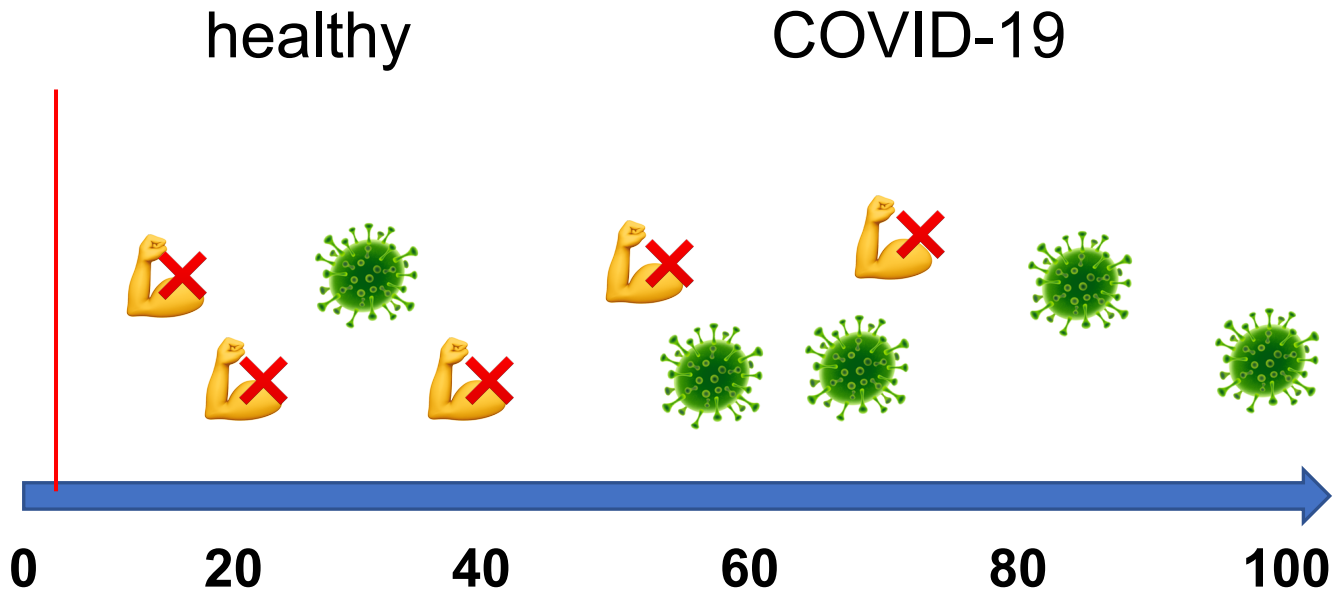
False positive rate
(FPR) = 2 / 5 = 0.4

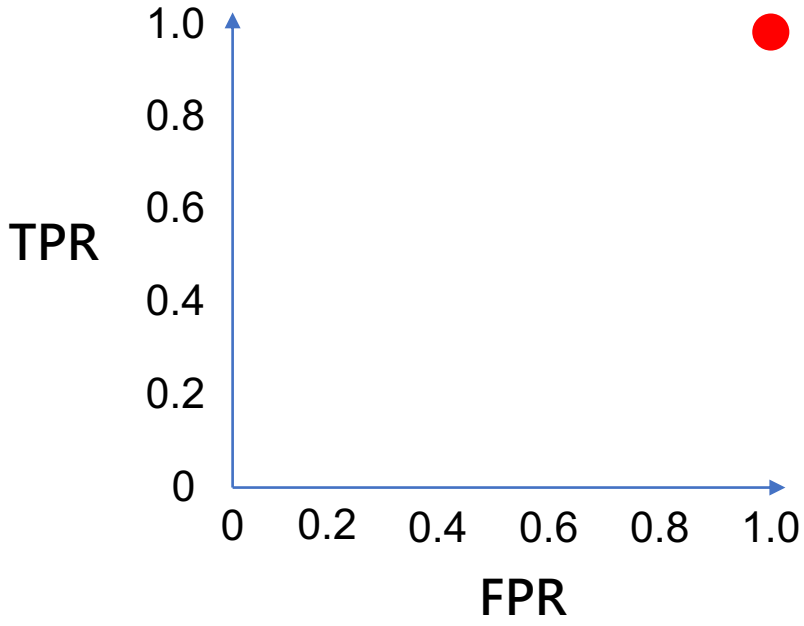


ROC Curve

假設我們用模型對10個人做COVID-19檢測：
(healthy 代表 negative; COVID-19 代表 positive)

TPR: 1.0, FPR: 1.0



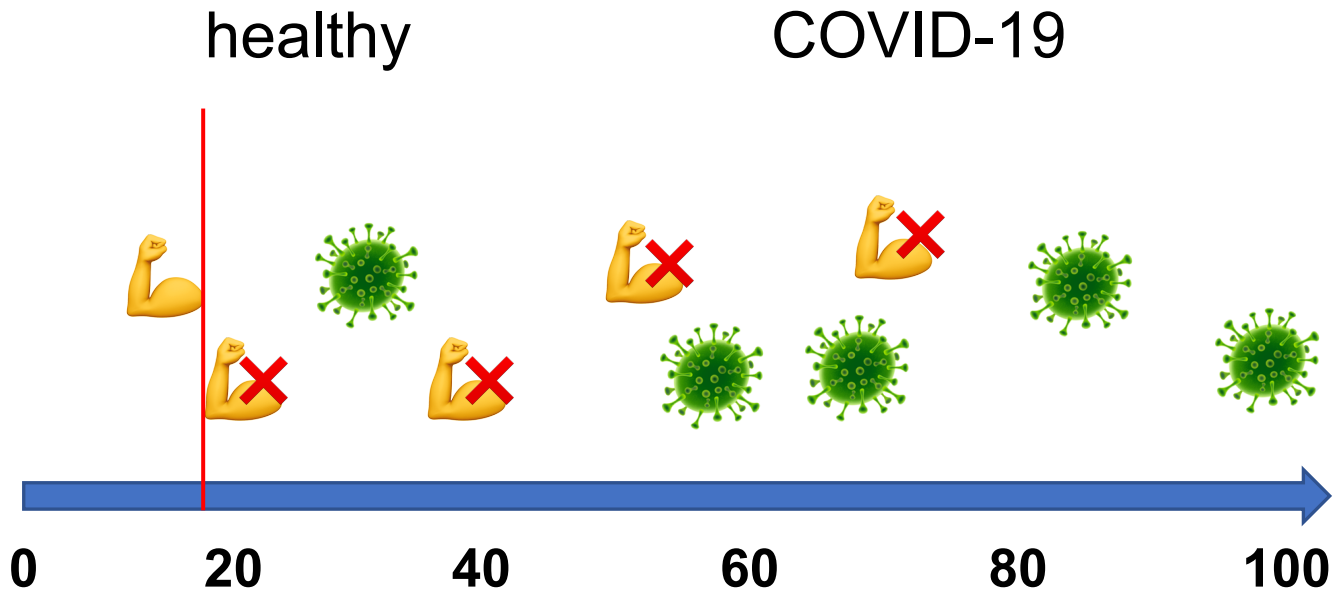


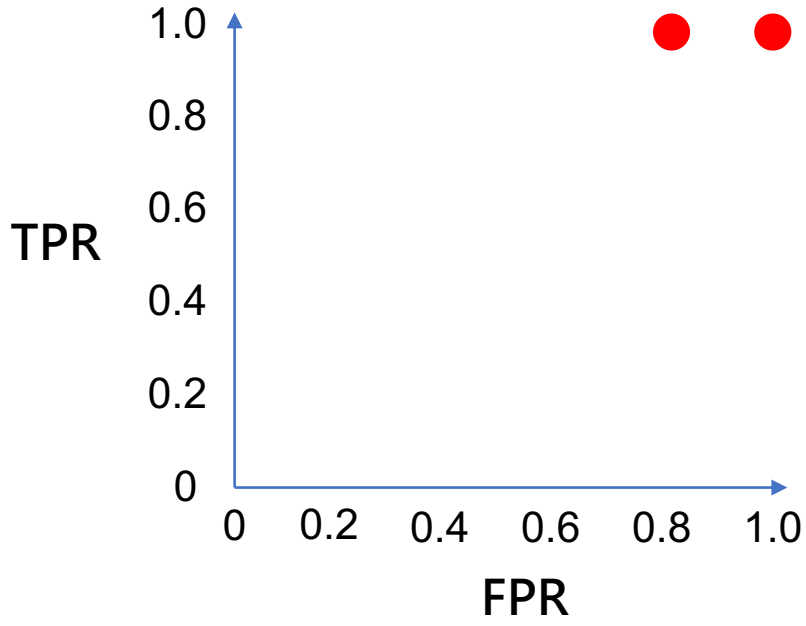
模型預測為疾病
的機率值 (%)

ROC Curve

假設我們用模型對10個人做COVID-19檢測：
(healthy 代表 negative; COVID-19 代表 positive)

TPR: 1.0, FPR: 0.8



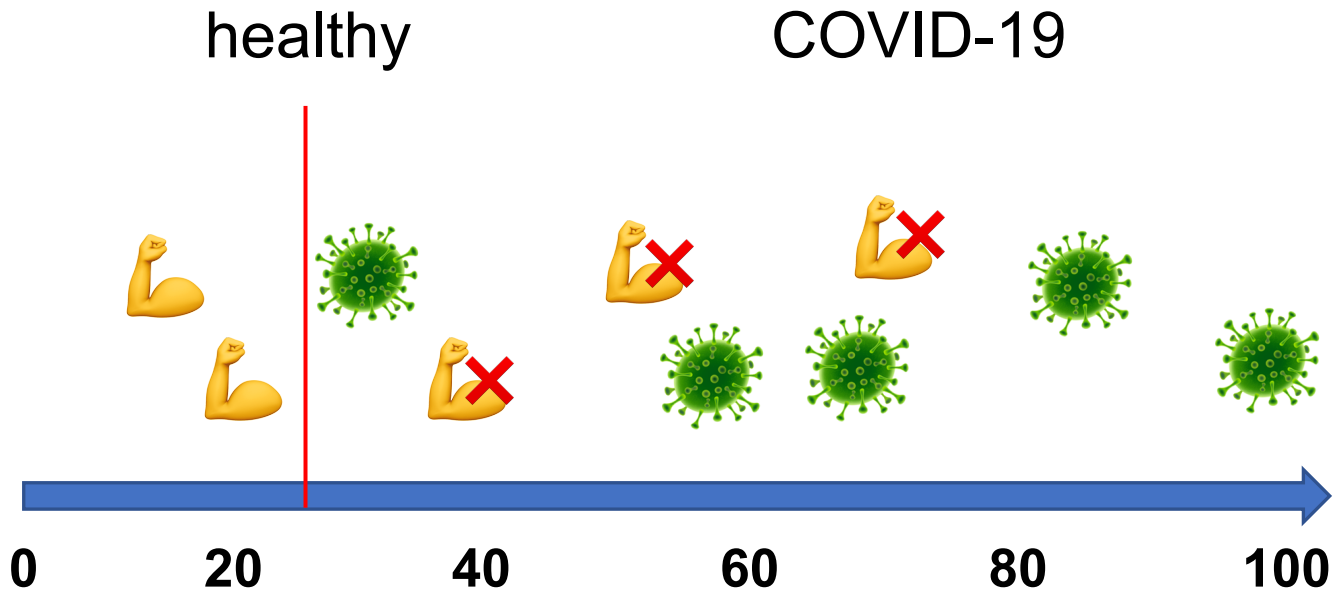


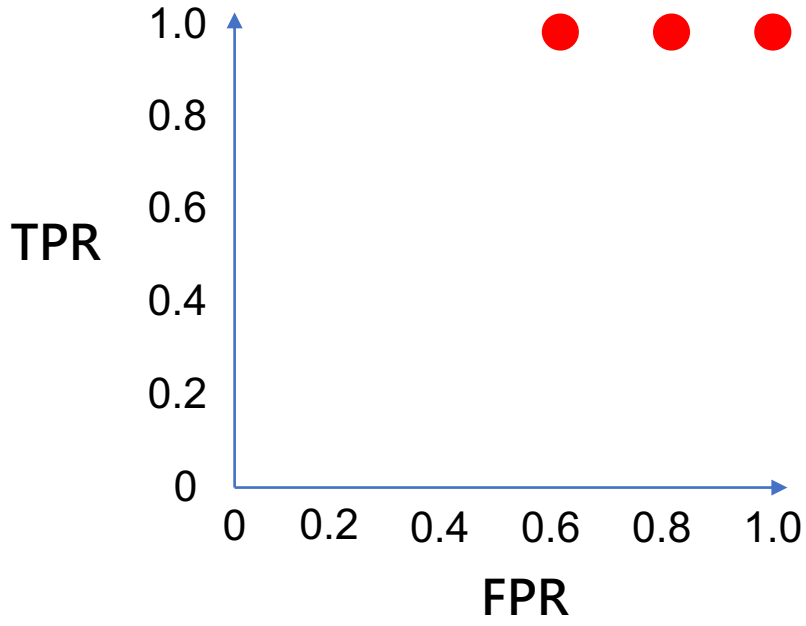
模型預測為疾病
的機率值 (%)

ROC Curve

假設我們用模型對10個人做COVID-19檢測：
(healthy 代表 negative; COVID-19 代表 positive)

TPR: 1.0, FPR: 0.6



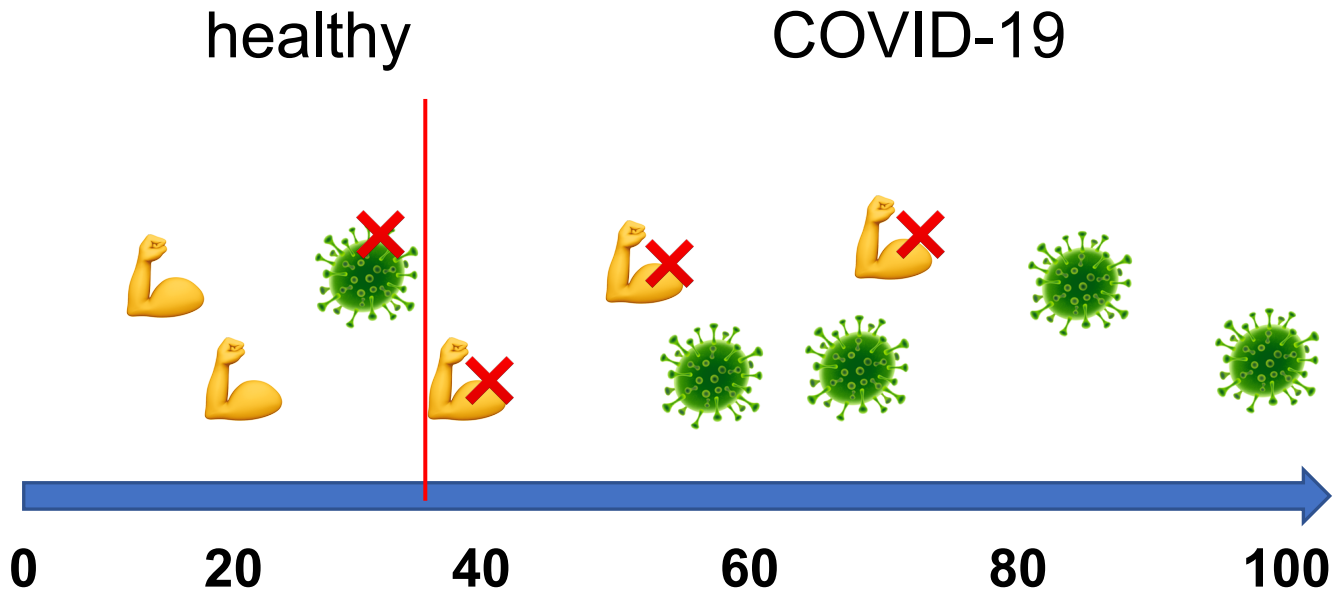


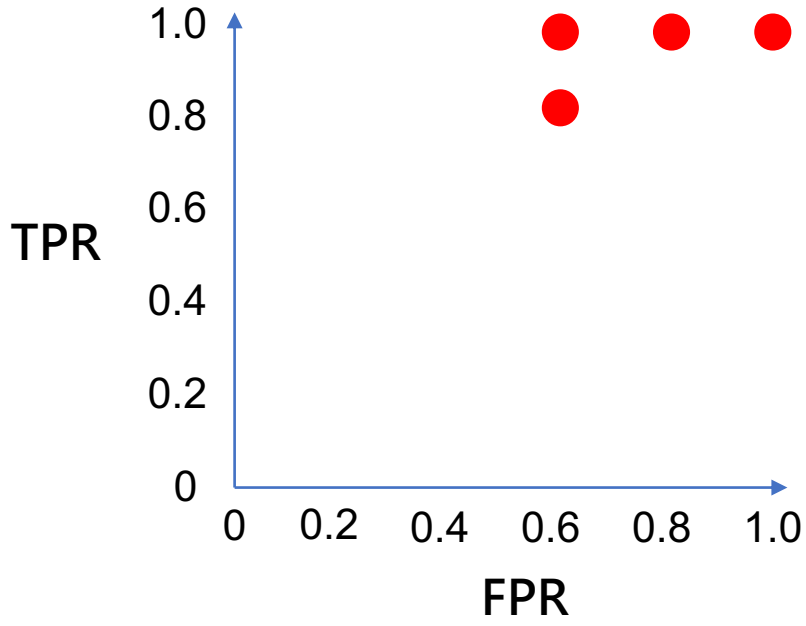
模型預測為疾病
的機率值 (%)

ROC Curve

假設我們用模型對10個人做COVID-19檢測：
(healthy 代表 negative; COVID-19 代表 positive)

TPR: 0.8, FPR: 0.6



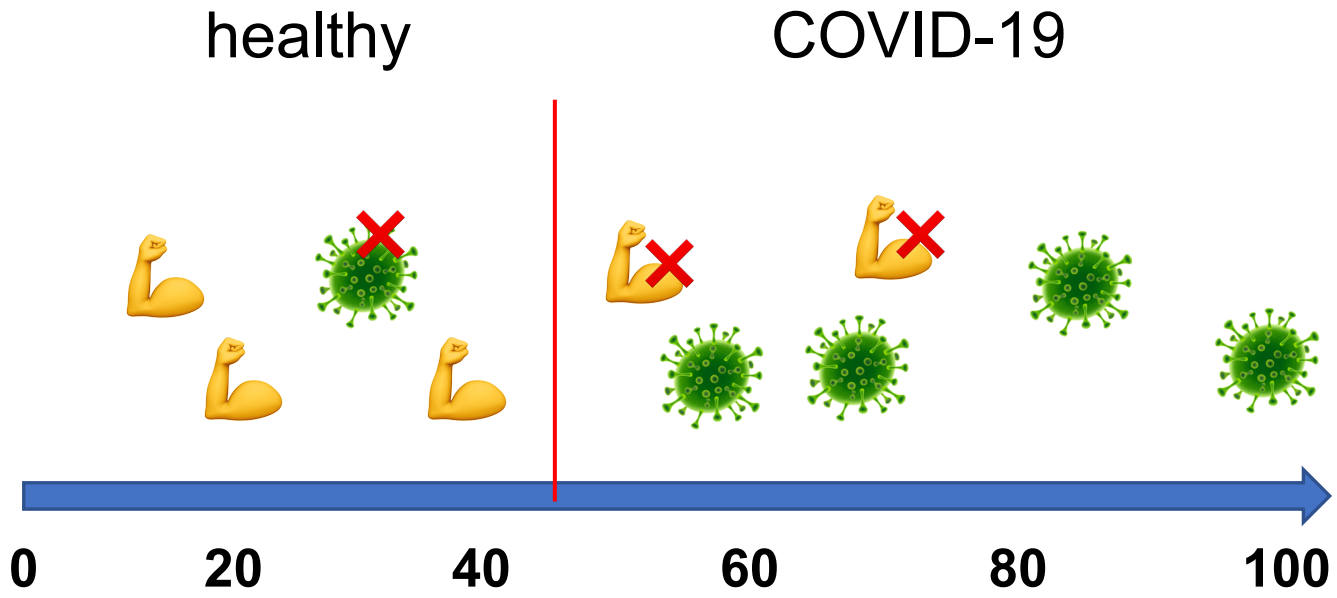


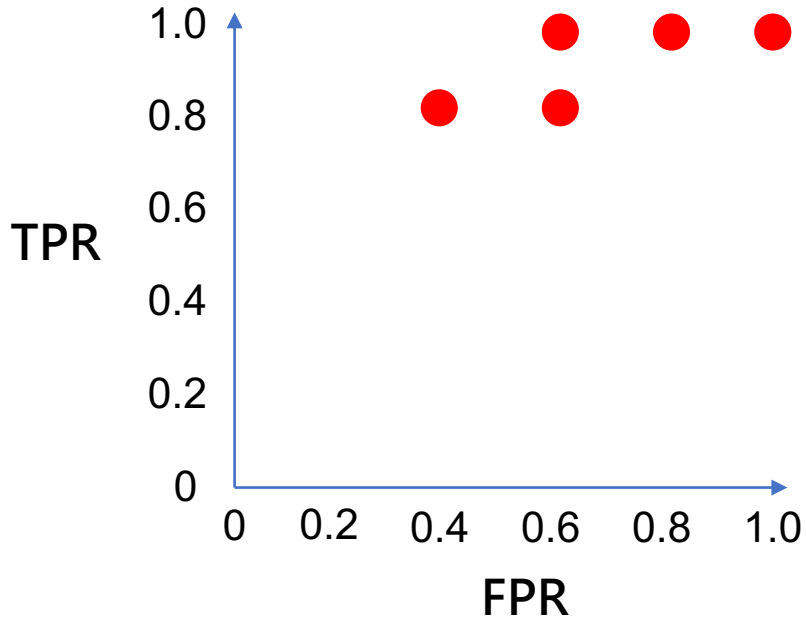
模型預測為疾病
的機率值 (%)

ROC Curve

假設我們用模型對10個人做COVID-19檢測：
(healthy 代表 negative; COVID-19 代表 positive)

TPR: 0.8, FPR: 0.4



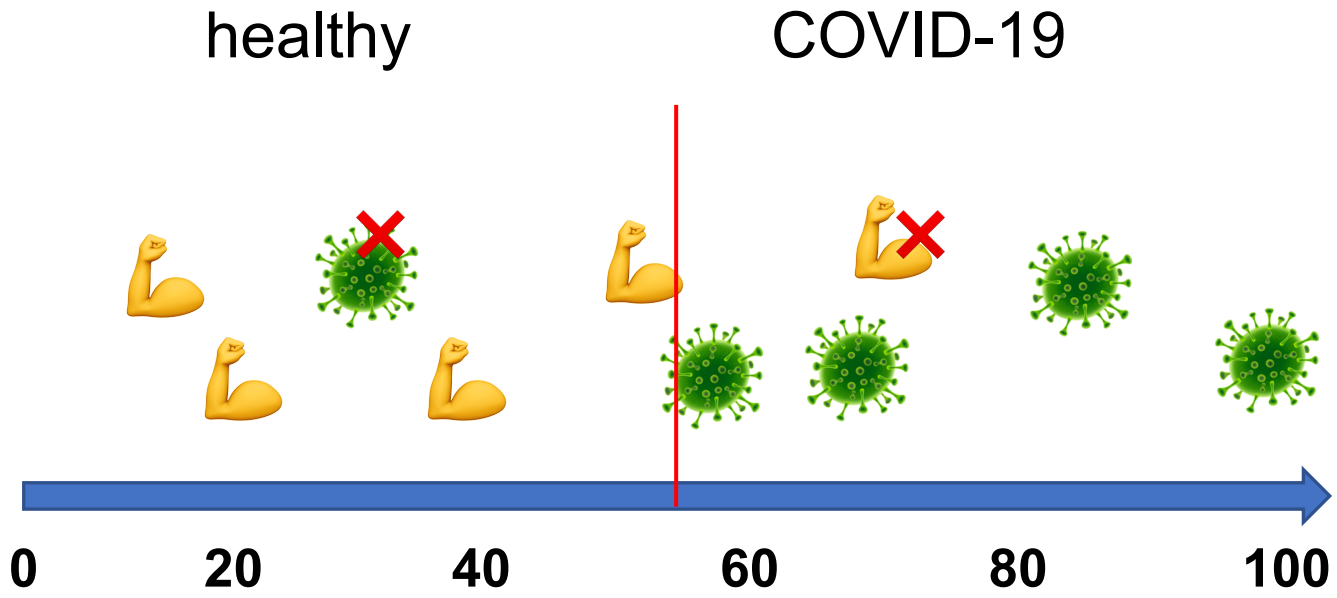


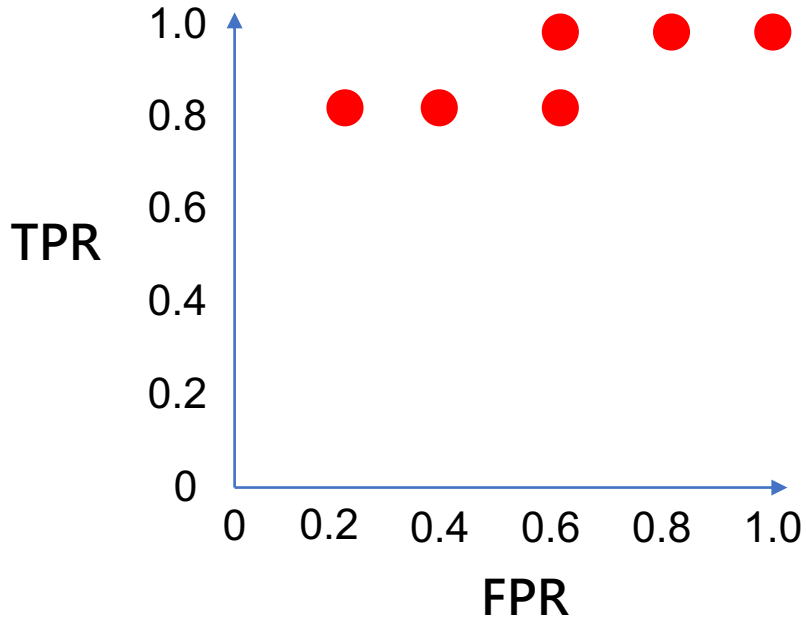
模型預測為疾病
的機率值 (%)

ROC Curve

假設我們用模型對10個人做COVID-19檢測：
(healthy 代表 negative; COVID-19 代表 positive)

TPR: 0.8, FPR: 0.2



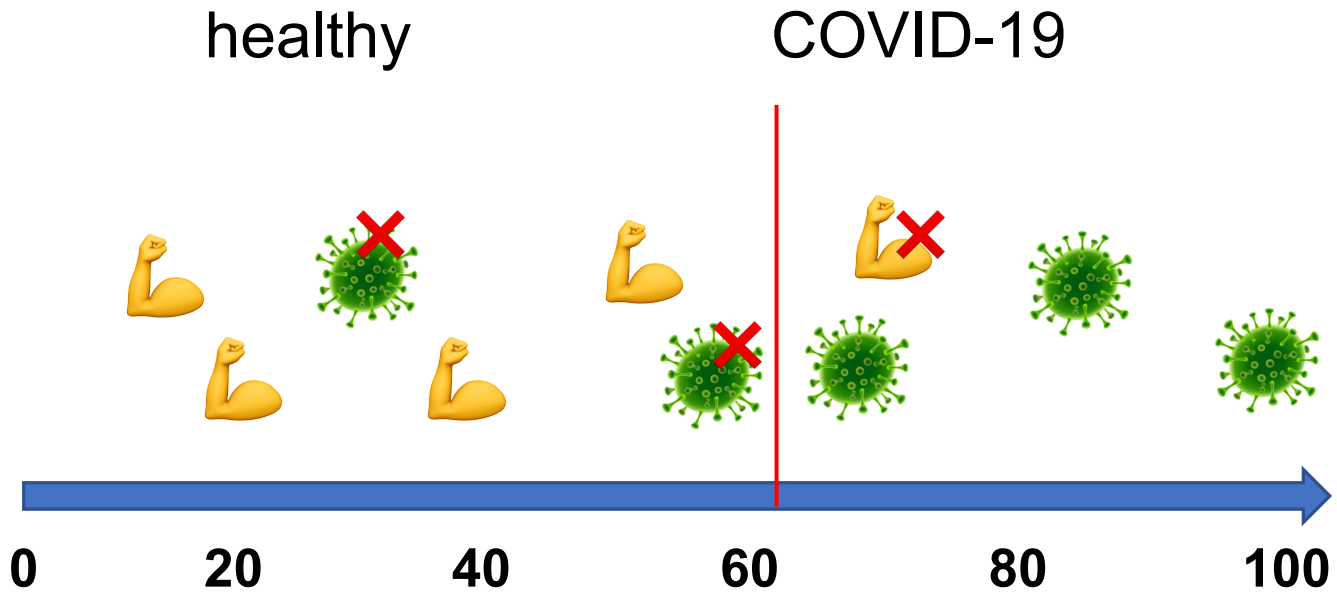


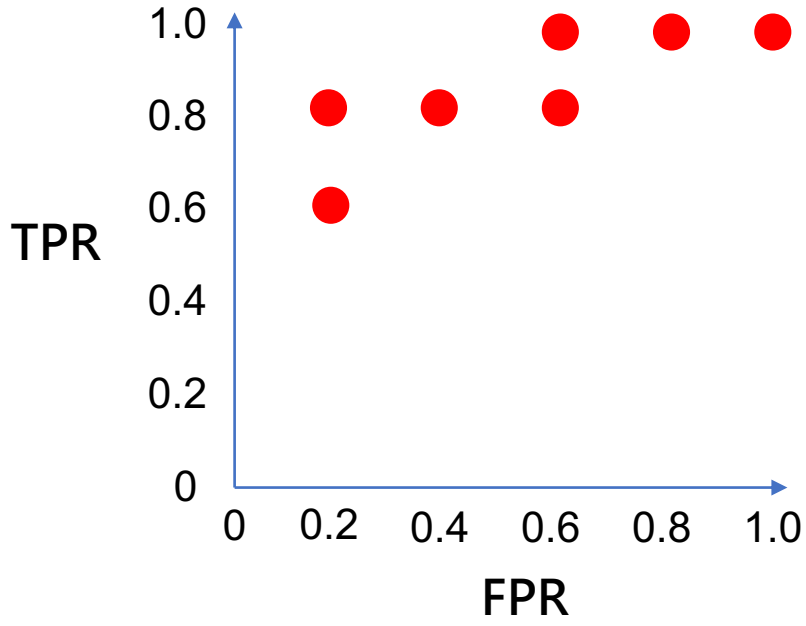
模型預測為疾病
的機率值 (%)

ROC Curve

假設我們用模型對10個人做COVID-19檢測：
(healthy 代表 negative; COVID-19 代表 positive)

TPR: 0.6, FPR: 0.2



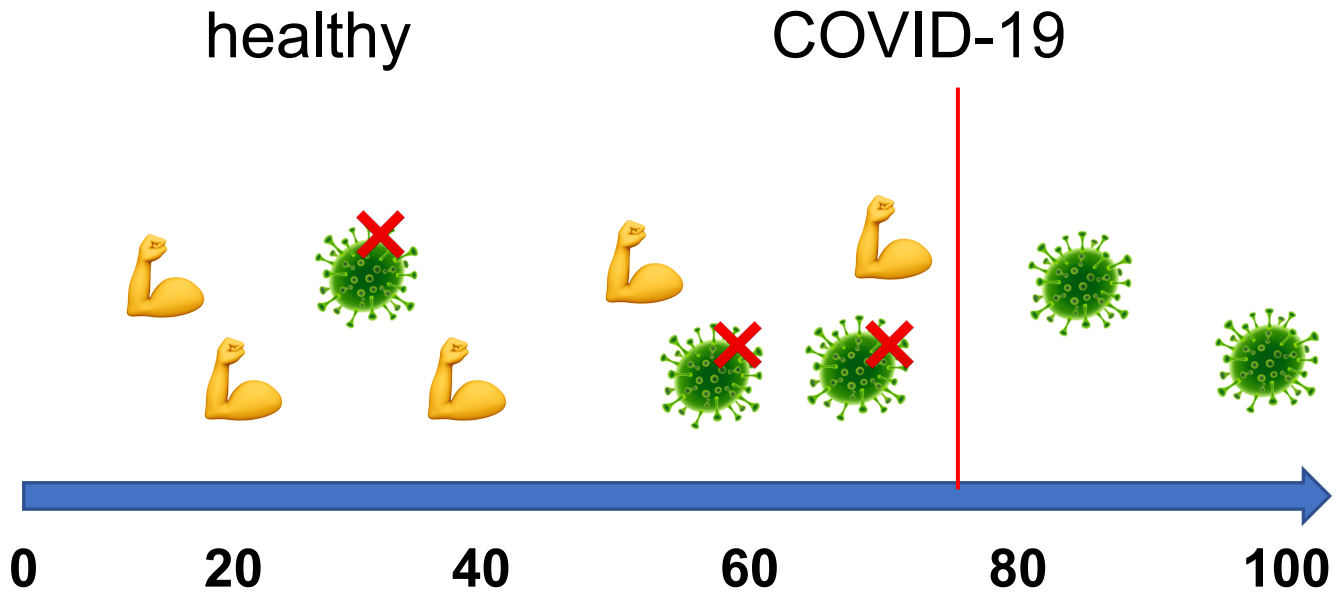


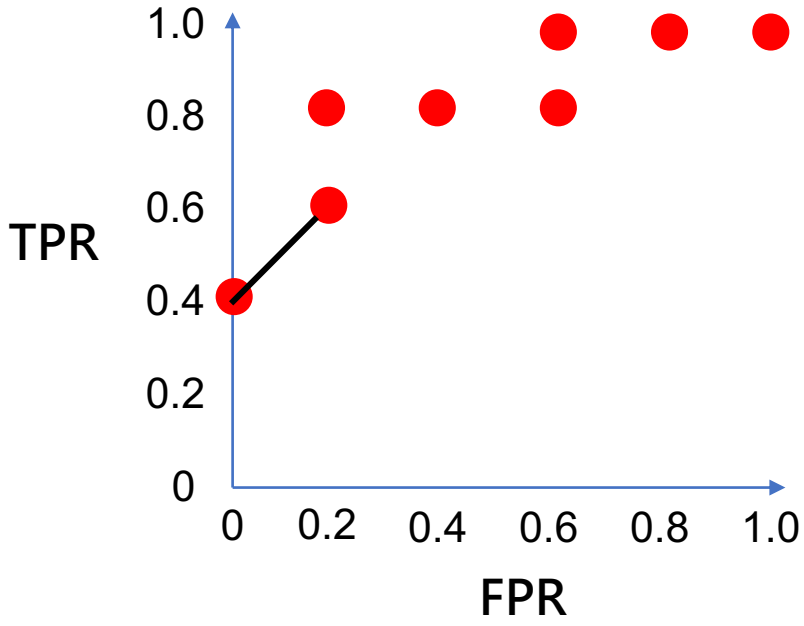
模型預測為疾病
的機率值 (%)

ROC Curve

假設我們用模型對10個人做COVID-19檢測：
(healthy 代表 negative; COVID-19 代表 positive)

TPR: 0.4, FPR: 0.0



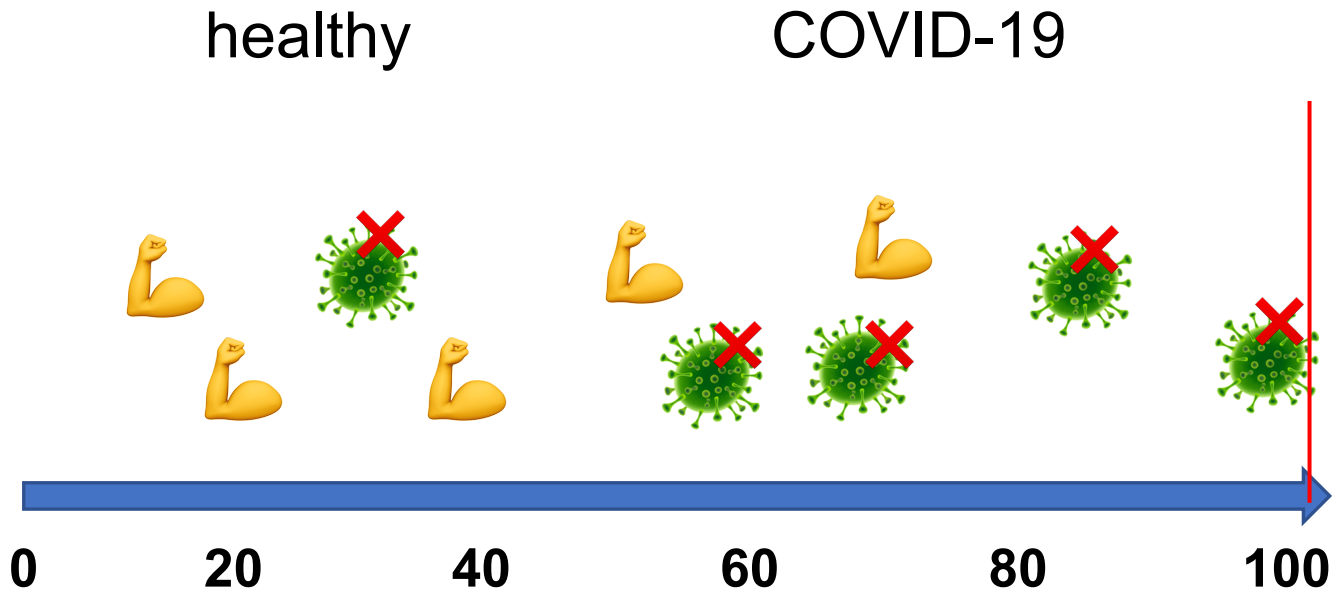


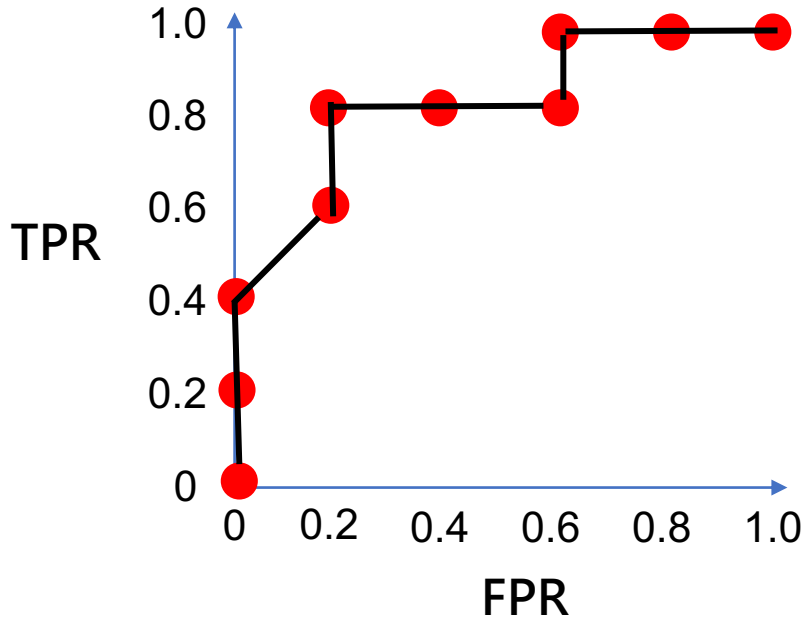
模型預測為疾病
的機率值 (%)

ROC Curve

假設我們用模型對10個人做COVID-19檢測：
(healthy 代表 negative; COVID-19 代表 positive)

TPR: 0.0, FPR: 0.0



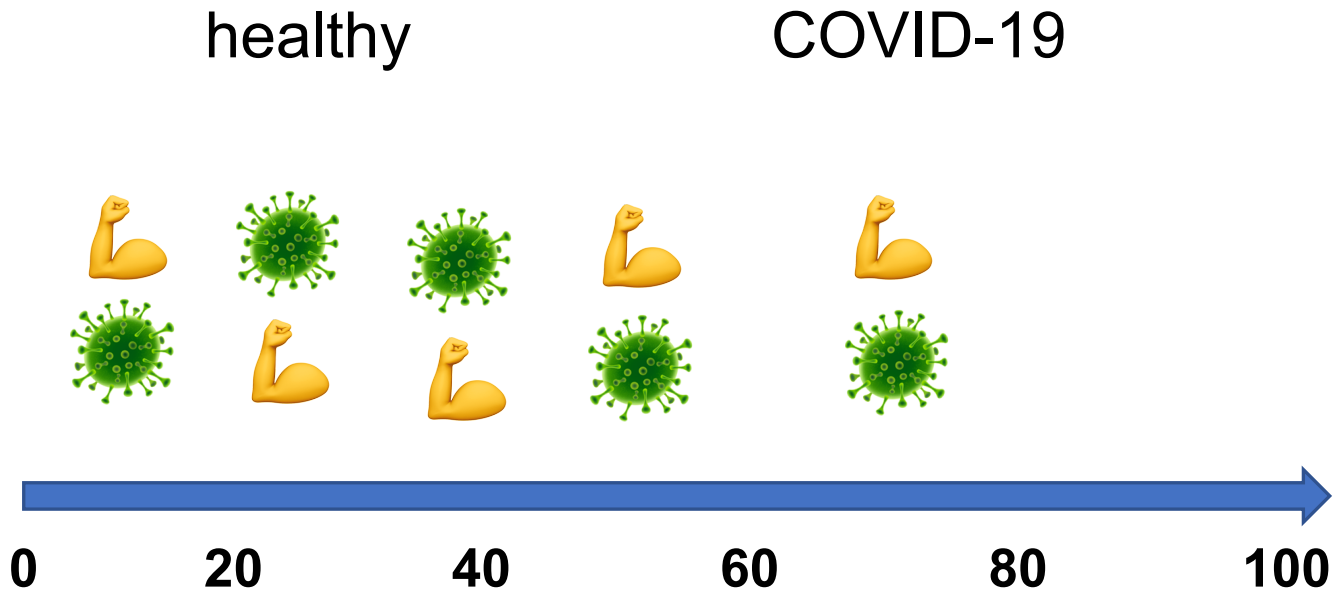


模型預測為疾病
的機率值 (%)

ROC Curve (bad case)

(Useful!!) https://developers.google.com/machine-learning/crash-course/classification/roc-and-auc?hl=zh-tw

Bad case: 模型對於 positive 和 negative 區分能力差



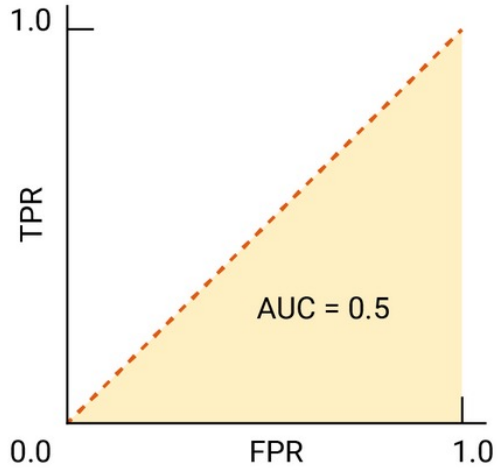


圖 2 ROC 和 AUC 完全隨機猜測。

完美模型的條件

- 分類結果完全正確
  - 沒有誤判
    - No False Negatives: TPR 越高越好
    - No False Positives: FPR 越低越好
- 完美模型中存在一個「理想 threshold 區間」，使我們能同時得到 TPR = 1 且 FPR = 0

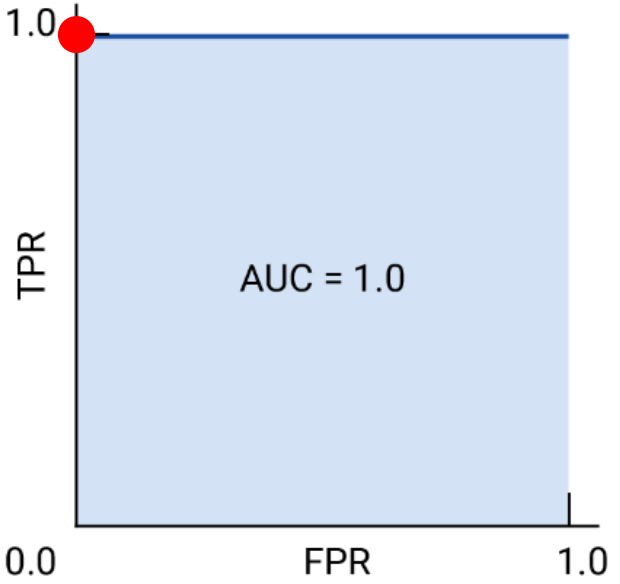


圖 1：ROC 和 AUC，這是一個完美假設的模型。

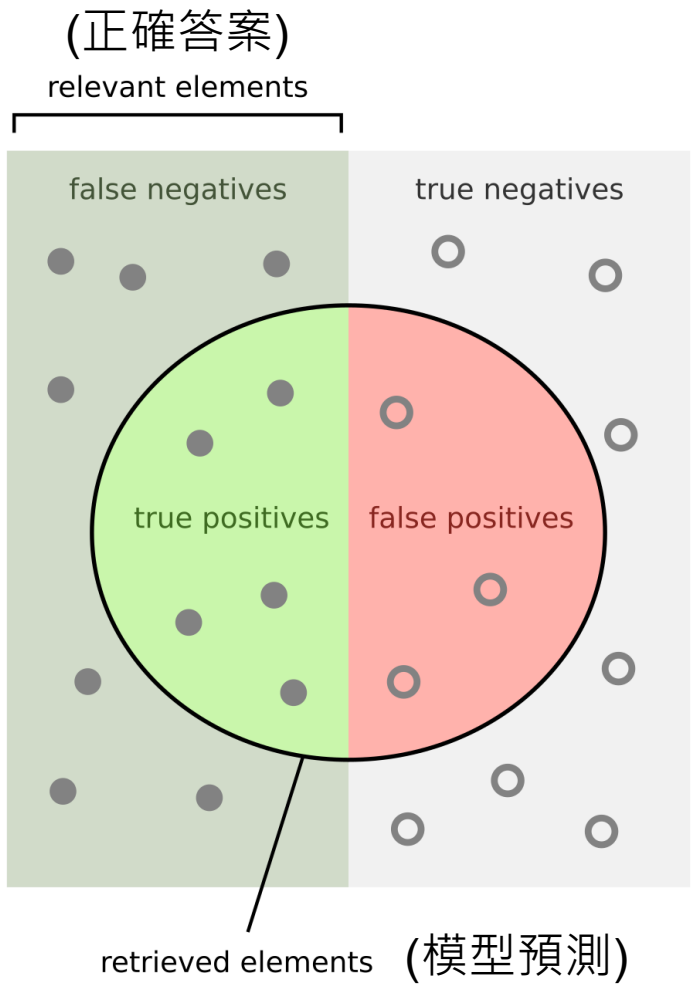
Confusion Matrix (混淆矩陣) for ROC

	Actually positive	**Actually negative**
Predicted positive	True positive (TP)	False positive (FP)
Predicted negative	False negative (FN)	True negative (TN)

- TP / (TP + FN): True Positive Rate (TPR)
  - 又稱作 Recall
- FP / (FP+TN): False Positive Rate (FPR)



Recall and Precision



How many retrieved items are relevant?

$$\text{Precision} = \frac{\text{true positives}}{\text{retrieved elements}}$$

How many relevant items are retrieved?

$$\text{Recall} = \frac{\text{true positives}}{\text{relevant elements}}$$

https://en.wikipedia.org/wiki/Precision_and_recall

Confusion Matrix for General Uses

	Actually positive	**Actually negative**
Predicted positive	True positive (TP) ↑	False positive (FP)
Predicted negative	False negative (FN)	True negative (TN) ↑

- 斜對角數值越大越好
  - TP 和 TN 越大越好
- Precision = TP / (TP + FP)
  - 模型預測的TP比例
- Recall = TP / (TP + FN)
- F1-score (廣義) = 2(Precision*Recall) / (Precision+Recall)



roc_curve using scikit-learn

- https://scikit-learn.org/stable/modules/generated/sklearn.metrics.roc_curve.html

```
import numpy as np
from sklearn import metrics

y = np.array([1, 1, 2, 2])
scores = np.array([0.1, 0.4, 0.35, 0.8])
fpr, tpr, thresholds = metrics.roc_curve(y, scores, pos_label=2)
```

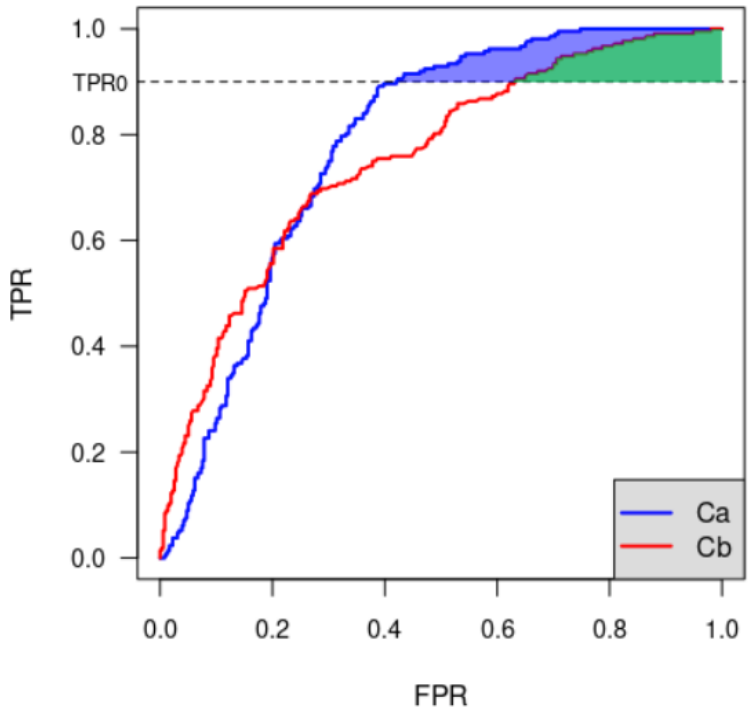


Partial AUC (pAUC)

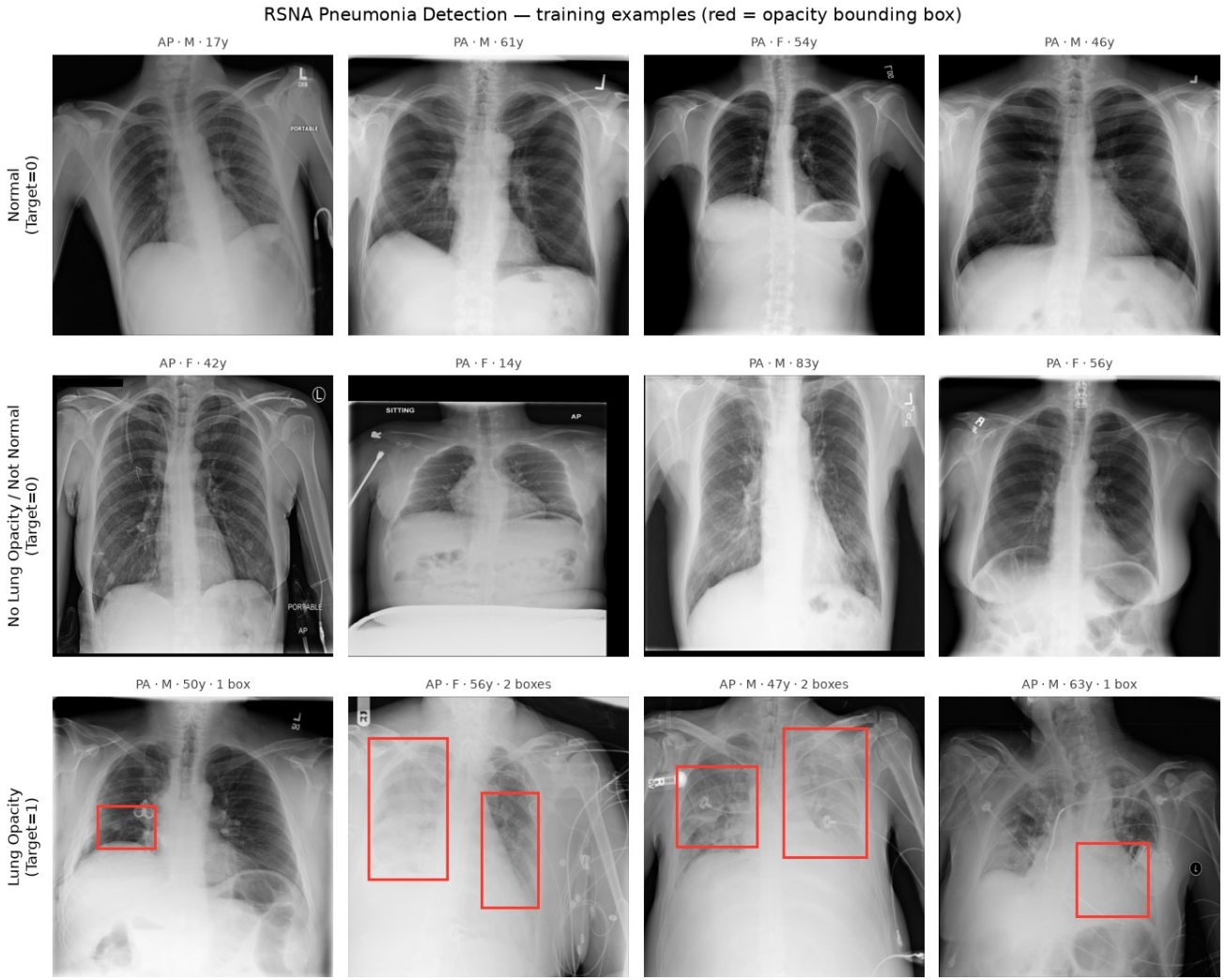
Submissions are evaluated on **partial area under the ROC curve (pAUC)** above 80% true positive rate (TPR) for binary classification of malignant examples. (See the implementation in the notebook ISIC pAUC-aboveTPR.)

The receiver operating characteristic (ROC) curve illustrates the diagnostic ability of a given binary classifier system as its discrimination threshold is varied. However, there are regions in the ROC space where the values of TPR are unacceptable in clinical practice. Systems that aid in diagnosing cancers are required to be highly-sensitive, so this metric focuses on the area under the ROC curve AND above 80% TRP. Hence, scores range from [0.0, 0.2].

The shaded regions in the following example represents the pAUC of two arbitrary algorithms (Ca and Cb) at an



RSNA Pneumonia Detection



RSNA Pneumonia – training data

stage_2_train_labels.csv (1.49 MB)

Detail Compact Column 6 of 6 columns

# x	# y	# width	# height	# Target
2 – 835	2 – 881	40 – 528	45 – 942	0 – 1
				0
				0
				0
				0
264.0	152.0	213.0	379.0	1
562.0	152.0	256.0	453.0	1

Data Explorer

3.96 GB

- stage_2_test_images
- stage_2_train_images
- GCP Credits Request Link
- stage_2_detailed_class_in
- stage_2_sample_submissi
- stage_2_train_labels.csv

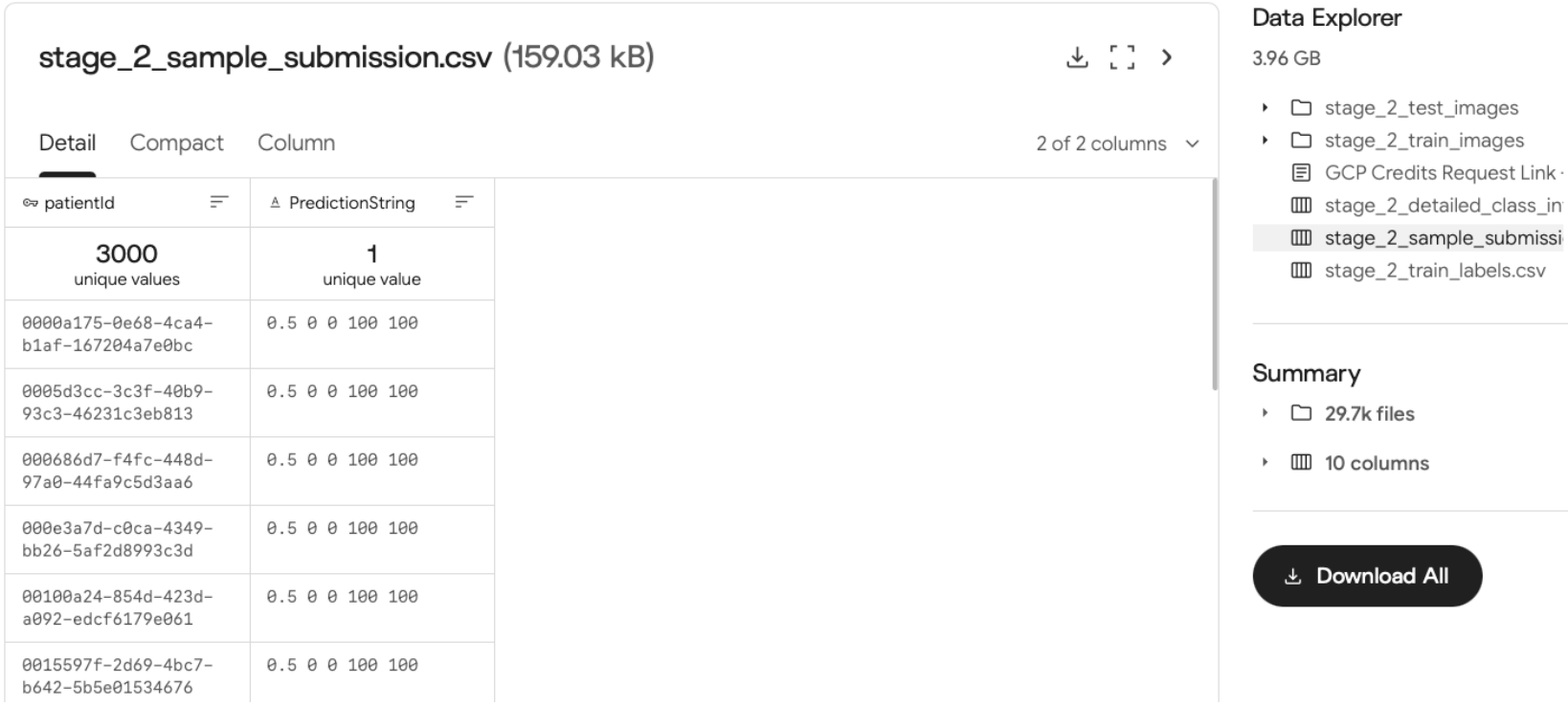
Summary

- 29.7k files
- 10 columns

Download All



RSNA Pneumonia – 上傳內容



NOTE: In nearly all cases confidence will have **no** impact on scoring. It exists primarily to allow for submission boxes to be evaluated in a particular order to resolve extreme edge cases. None of these edge cases are known to exist in the data set. If you do not wish to use or calculate confidence you can use a placeholder value - like 1.0 - to indicate that no particular order applies to the evaluation of your submission boxes.

評估指標：IOU (Intersection over Union)

$$Intersection\ over\ Union\ (IoU) = \frac{Area\ of\ Overlap}{Area\ of\ Union}$$

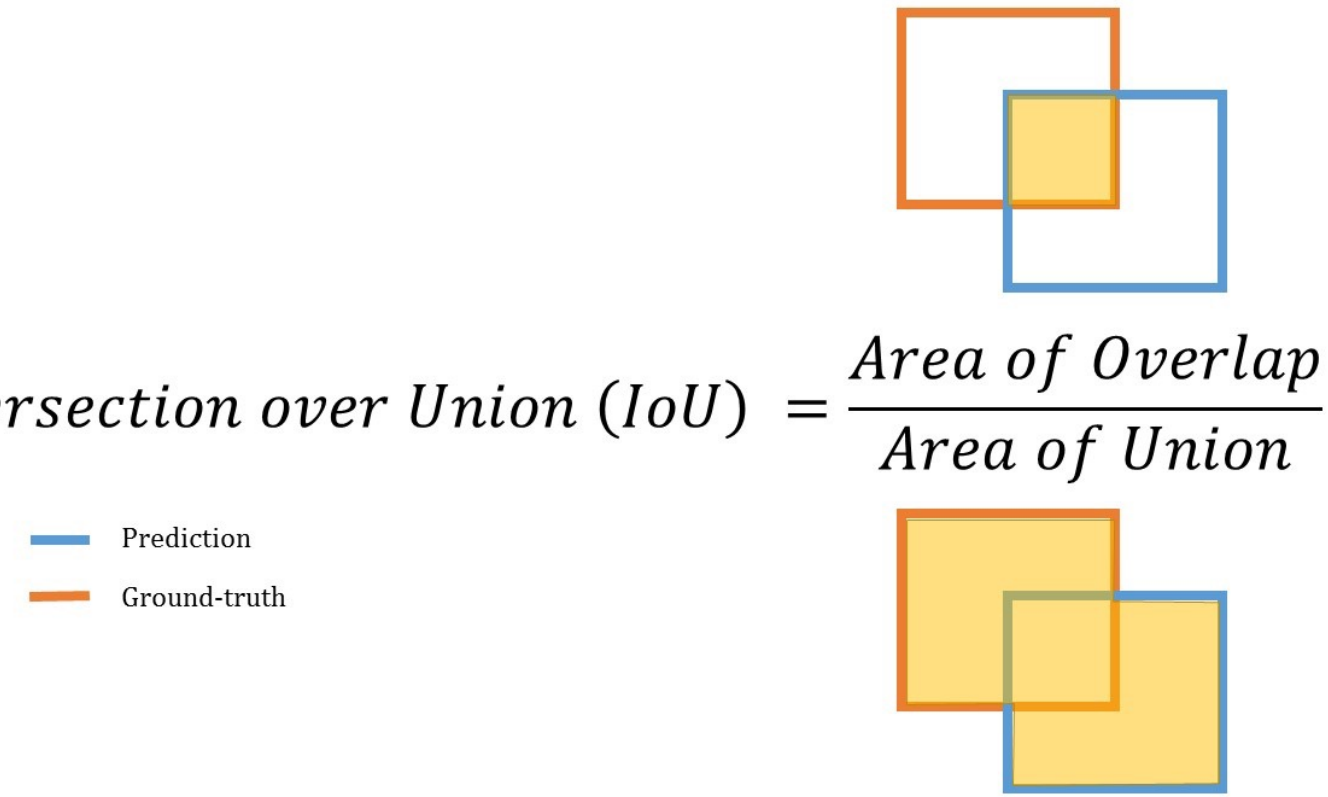
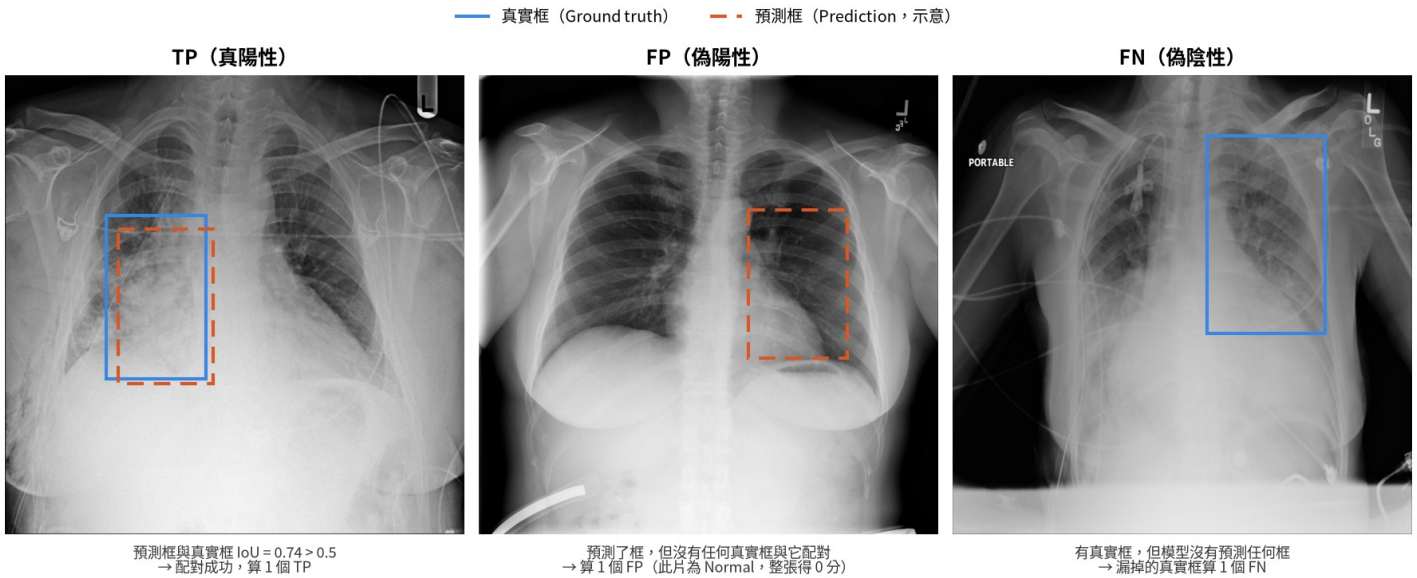


Figure source: https://www.kaggle.com/competitions/rsna-pneumonia-detection-challenge/overview

RSNA Pneumonia – 評估指標



The metric sweeps over a range of IoU thresholds, at each point calculating an average precision value. The threshold values range from 0.4 to 0.75 with a step size of 0.05: `(0.4, 0.45, 0.5, 0.55, 0.6, 0.65, 0.7, 0.75)`. In other words, at a threshold of 0.5, a predicted object is considered a "hit" if its intersection over union with a ground truth object is greater than 0.5.

At each threshold value t, a precision value is calculated based on the number of true positives (TP), false negatives (FN), and false positives (FP) resulting from comparing the predicted object to all ground truth objects:

$$\frac{TP(t)}{TP(t) + FP(t) + FN(t)}.$$

A true positive is counted when a single predicted object matches a ground truth object with an IoU above the threshold. A false positive indicates a predicted object had no associated ground truth object. A false negative indicates a ground truth object had no associated predicted object.

Important note: if there are no ground truth objects at all for a given image, ANY number of predictions (false positives) will result in the image receiving a score of zero, and being included in the mean average precision.

The average precision of a single image is calculated as the mean of the above precision values at each IoU threshold:

$$\frac{1}{|thresholds|} \sum_t \frac{TP(t)}{TP(t) + FP(t) + FN(t)}.$$

https://yy-programer.blogspot.com/2020/06/iouapmap.html

Name That LLM – 資料（不一定訓練）

train.csv (2.14 MB)

Detail | Compact | Column | 4 of 4 columns

id	prompt_id	text	model
2699 unique values	150 unique values	2699 unique values	qwen3.8-max 6% claude-sonnet-5 6% Other (2399) 89%
0018b3765972	p081	When a language dies, a particular astronomy dims: not only names for stars, but whole grammars of a...	qwen3.8-max
0041ac86ee93	p073	Data science communities matter because they accelerate learning through shared knowledge, code, and...	claude-sonnet-5
005a19083eae	p120	In the final twilight of time, the last embers of ancient suns falter into silent rust, leaving behi...	gemini-3.6-flash
009319596f10	p073	Data science communities matter because they turn a fast-moving, technically	gpt-5.6-terra

Data Explorer

2.14 MB

- sample_submission.parquet
- test.csv
- train.csv

Summary

- 3 files
- 6 columns

Download All



Name That LLM – 上傳內容

test.csv (2.22 kB)

Detail | Compact | Column | 2 of 2 columns

id	text
3 unique values	3 unique values
00353fb1c541	The more efficiently we use a resource, the more of it we ultimately consume: cheaper energy from ef...
005f5a9a6139	The Stern–Gerlach experiment, performed in 1922 by Otto Stern and Walther Gerlach, sent a narrow bea...
0062a53d5487	Hundreds of millions of years ago, fluctuating climates and tectonic shifts isolated massive landloc...

No more data to show

Data Explorer

2.14 MB

- sample_submission.parquet
- test.csv
- train.csv

Submission File

This is a Code Competition. Your notebook reads the test set from `/kaggle/input/competitions/name-that-llm/test.csv` and must write a file named `submission.parquet` to the notebook's working directory.

During an interactive session `test.csv` contains a single example row so your notebook can run end to end. When your notebook is submitted for scoring, it is re-run against the full hidden test set (~4000 passages).

For each `id` in the test set, predict the authoring `model`. Every `id` must appear exactly once. The file should contain a header and have the following format:

```
id,model
a1b2c3d4e5f6,gpt-6-astra
b2c3d4e5f6a7,claude-opus-5
c3d4e5f6a7b8,gemini-3.8-flash
...
```



評估指標：18 個模型的平均 Acc.

Submissions are evaluated on **balanced accuracy**: the average, over all candidate models, of the fraction of that model's test passages you labelled correctly (per-class recall).

$$\text{Balanced Accuracy} = \frac{1}{K}\sum_{k=1}^{K}\frac{|\,i : y_i = k \ \wedge\ \hat{y}_i = k\,|}{|\,i : y_i = k\,|}$$

where $K = 18$ is the number of candidate models, y_i is the true authoring model and $\hat{y}_i$ is your prediction for passage i.

The number of test passages per model is **not** equal, and the mix of models differs between the public and private leaderboard. Balanced accuracy gives every model the same weight.



Figure source: https://www.kaggle.com/competitions/rsna-pneumonia-detection-challenge/overview

SEMEVAL 2027 TASK 2

DICO NLI

DIRECTIONAL CONSISTENT NLI

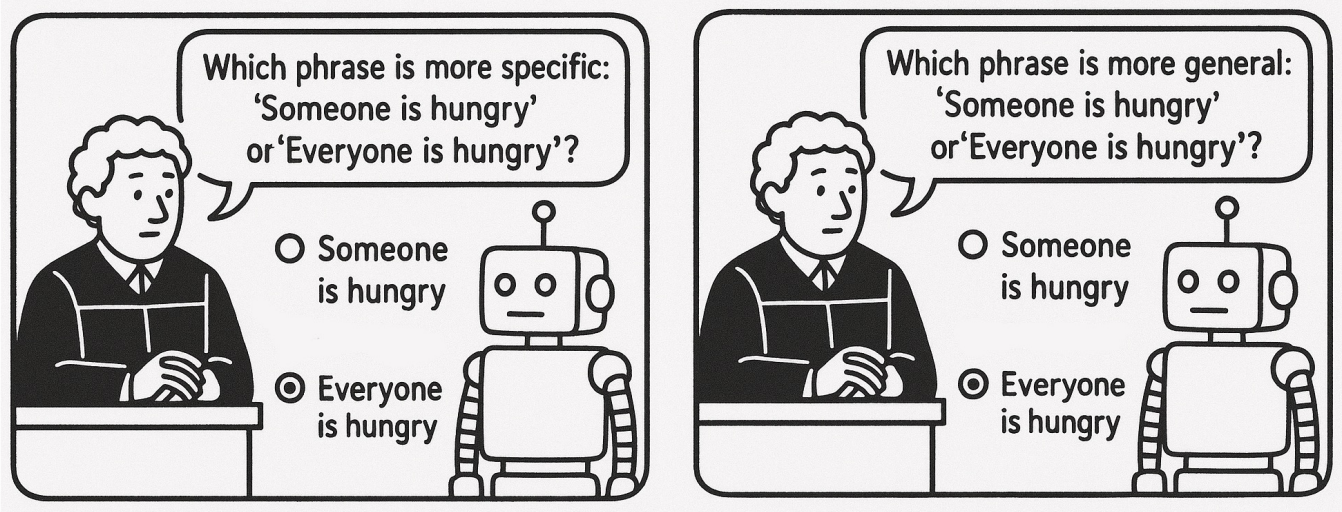


Figure source: https://inigolopezgazpio.net/SemEval-2027-Task-2-DiCo-NLI/

Natural Language Inference (NLI)

前提 (premise)	假設 (Hypothesis)	LABEL
Everyone is hungry.	Someone is hungry.	Entailment (蘊含)
Someone is hungry.	Everyone is hungry.	Neutral (中立)
Someone is hungry.	No one is hungry.	Contradiction (反對)



Directional Consistent NLI (Dico-NLI)

前提 (premise)	假設 (Hypothesis)	**NLI LABEL**	**Dico-NLI LABEL**
Everyone is hungry.	Someone is hungry.	Entailment (蘊含)	Forward Entailment
Someone is hungry.	Everyone is hungry.	Neutral (中立)	Backward Entailment
Someone is hungry.	No one is hungry.	Contradiction (反對)	-
Someone is hungry.	At least one person is hungry.	Contradiction (反對)	Equivalence



Dico-NLI Dataset Statistics

Current public train/dev instance counts are:

Split	Track 1 English	Track 2 Spanish	Track 3 Basque	Track 4 Mixed multilingual
Train	3042	3042	3042	18252
Dev	660	660	660	3960

Current label counts are:

Split	Track(s)	BACKWARD_ENTAILMENT	EQUIVALENCE	FORWARD_ENTAILMENT
Train	1/2/3	881	802	881
Train	4	5286	4812	5286
Dev	1/2/3	190	174	190
Dev	4	1140	1044	1140



評估指標

Metric	What is measues
Weighted F1	sklearn.metrics.f1_score(y_true, y_pred, average="weighted")
SoftCons	Input 兩句前後交換，模型是否會輸出相反的 labels (無論答案是否正確)
HardCons	Input 兩句前後交換，模型是否會輸出相反的 labels (答案必須正確)



Examples for SoftCons and HardCons

SoftCons

Premise	Hypothesis	Prediction
Everyone is hungry	Someone is hungry	FORWARD_ENTAILMENT
Someone is hungry	Everyone is hungry	BACKWARD_ENTAILMENT

HardCons

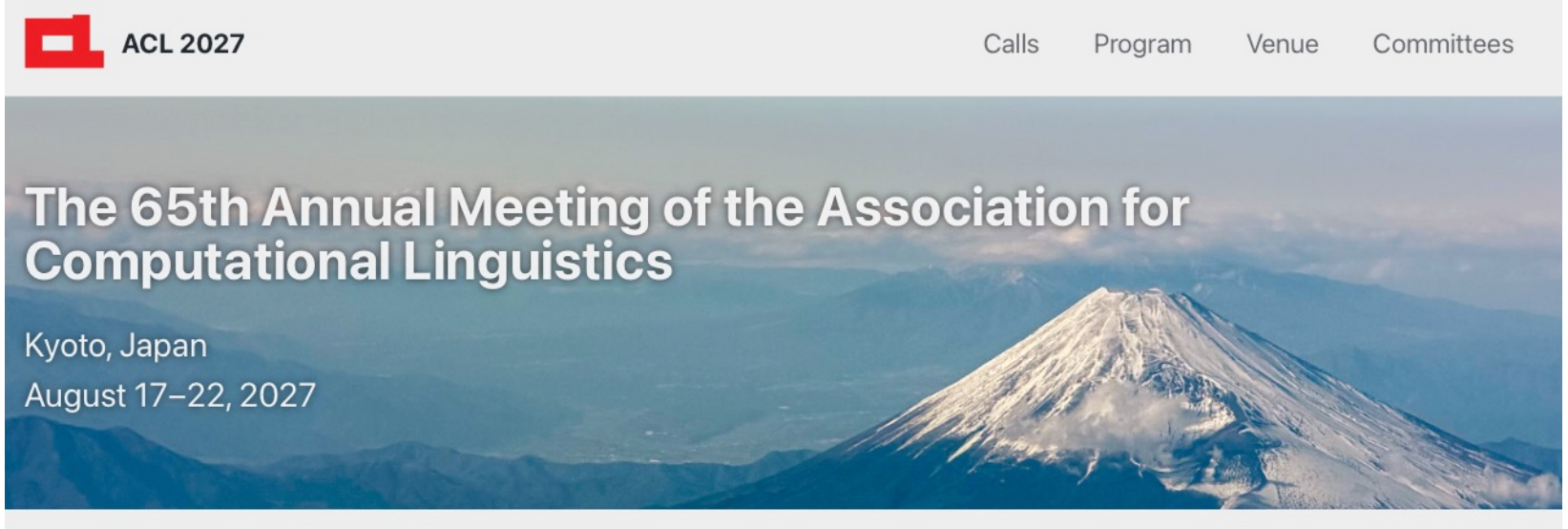
Premise	Hypothesis	Gold label	Prediction
Everyone is hungry	Someone is hungry	FORWARD_ENTAILMENT	FORWARD_ENTAILMENT
Someone is hungry	Everyone is hungry	BACKWARD_ENTAILMENT	BACKWARD_ENTAILMENT



官方範例程式碼以及上傳平台

https://inigolopezgazpio.net/SemEval-2027-Task-2-DiCo-NLI/starter_kit/

https://www.codabench.org/competitions/18038/





Thank you!

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TA: 鄭茹云

✉ m1461010@cgu.edu.tw